MODULE THREE

TRANSMISSION MODES, MEDIA AND ORGANIZATION

3.1 TRANSMISSION MODES

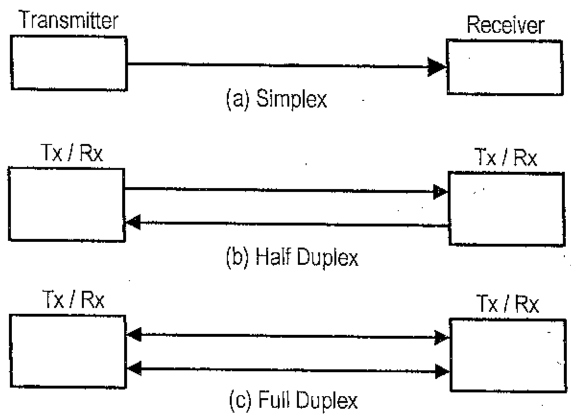
Data transmission mode refers to the techniques utilized to transmit data from one point to another. Three techniques can be used for data transmission, these are based on:

(1) Channel Type
(2) Serial/Parallel Transmission
(3) Asynchronous/Synchronous Transmission

CHANNEL TYPE:

Irrespective of the direction of data transfer, there are three types of transmission channels used to exchange information.

(i) Simplex
(ii) Half Duplex
(iii) Full Duplex



Simplex: - this mode refers to one-way transmission (only one direction) such as in radio or TV transmission. One party in the communication can send data to the other, but cannot receive data from the other end. It is usually not controlled by the transmission medium, but the nature of communication devices.

Half Duplex: - this is a two-way transmission (both direction); but in only one direction at a time (i.e. not simultaneously). Usually in this system, the receiver has to wait for the transmitter to finish transmitting before he can now transmit e.g. walkie-talkie. Communication between a computer and printer is half duplex.

Full Duplex: - this system allows transmission in both directions at the same time (simultaneously). Example, telephone to telephone communication or computer to computer communication and radar communication. A full duplex can be logically regarded as two half duplex operating in reverse direction. The benefits of this system include time saving and availability of full bandwidth.

SERIAL / PARALLEL MODE:

One major concern when transferring data from one device to another is the wiring, and of primary concern when considering the wiring is the data stream. Do we send 1 bit at a time; or do we group bits into larger groups and, if so how?

The digital information regardless of channel type can be classified, in terms of transmission mode, into serial or parallel transmission. Serial transmission means to transmit the data bit by bit with each clock tick, whereas parallel transmission means to transmit data byte by byte, word by word or even more with each clock tick.

SERIAL: - The information bits are transmitted over a communication circuit in a bit-by-bit fashion, that is one bit follows another, so we need only one communication channel (see figure 12), such as communication between a modem and a computer. Serial mode is the predominant method of transferring information between data communication equipments.

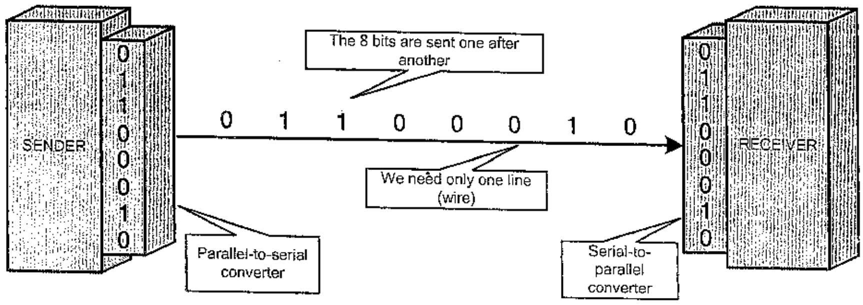


Fig.12: Serial transmission of an 8-bit code.

The advantage of serial transmission is that it uses only one communication channel, serial transmission reduces the cost of transmission over parallel transmission by a factor of n.

Since communication within devices is parallel, conversion devices are required at the interface between the sender and the line (parallel-to-serial) and between the line and the receiver (serial-to-parallel).

PARALLEL: - The bits are transmitted all at once (byte by byte). That is the binary data, consisting of 1s and 0s, may be organized into groups of n bits each such as the internal transfer of binary data within a computer or computer to printer communication. By grouping, we can send data n bits at a time instead of 1.

This method uses n wires to send n bits at one time. That way each bit has its own wire, and all n bits of one group can be transmitted with each clock tick from one device to another. Figure 13 shows how parallel transmission works for n = 8. Typically, the eight wires are bundled in a cable with a connector at each end.

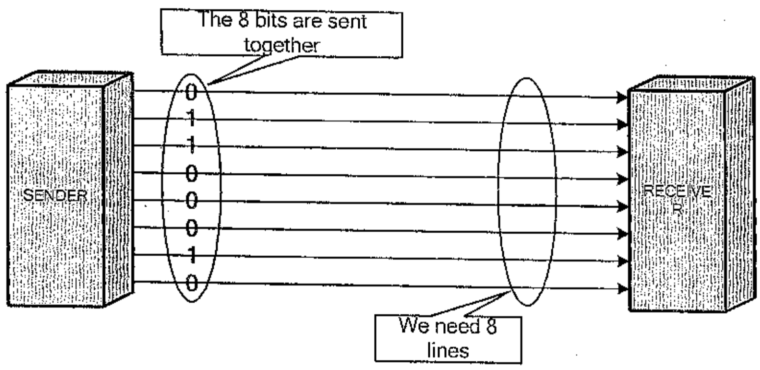


Fig.13: Parallel transmission of an 8-bit code.

The advantage of parallel transmission is speed. All else being equal, parallel transmission can increase the transfer speed by a factor of "n" over serial transmission. But there is a significant disadvantage, Cost. Parallel transmission requires "n" communication lines (wires) just to transmit the data stream. Because this is expensive, parallel transmission is usually limited to short distances.

ASYNCHRONOUS / SYNCHRONOUS TRANSMISSION:

Most data communication functions are performed by serial transmission. Two types of serial transmission mode are – (i) Asynchronous Transmission (ii) Synchronous Transmission

ASYNCHRONOUS TRANSMISSION: A transmission in which each information character is individually synchronized, usually by the use of a start and stop bit. The gap between each character is not a fixed length. It is often referred to as **start-stop transmission** because the transmitting device can transmit a character at any time it is convenient, and the receiving device will accept that character.

With asynchronous transmission each character is transmitted independently of all other characters. In order to separate the characters and synchronize transmission, a start bit (usually 0) and a stop bit (usually 1) are put on each end of the individual 8-bit character, so the total transmission is 10 bits per character, as shown in figure 14.

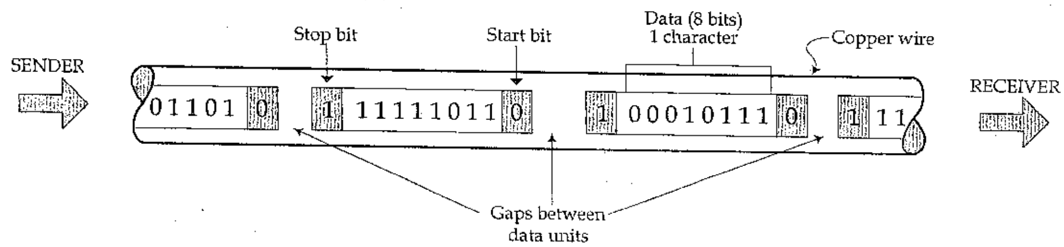


Fig.14: Asynchronous Transmission

The addition of start and stop bits and the insertion of gaps into the bit stream make asynchronous transmission slower than synchronous transmission. But it is cheap and effective, two advantages that make it an attractive choice for low-speed communication. A null modem is used to transmit asynchronously between two nearby computers.

SYNCHRONOUS TRANSMISSION: In this method of transmission, both the sending and receiving devices are synchronized with a common clock signal. This eliminates the need for the start and stop bits. The complete block of data is combined into longer "frames" which may contain multiple bytes or characters. Each byte in the frame however is transmitted with fixed time interval between the bits. Synchronous transmission is used for the high speed transmission of a block of characters, sometimes called a frame or packet.

Before sending the start of transmission, clocks at both ends are synchronized. This is achieved by sending special character bytes called "**Sync bytes**" between the sender and the receiver that lets the receiver know it is about to receive data. Included with these sync characters is the information about the rate of data transmission so that the receiving device can correctly adjust it's receive rate to match the transmission rate of the sending device.

After the sync characters, the transmitting device sends a long stream of data bits that may have thousands of bits. The receiving device, knowing what code is being used, counts off the appropriate number of bits for the first character, assumes this is the first character, and passes it to the computer. It then counts off the bits for the second character and so on.

If the sender is idle or does not transmit any character, the receiver searches for the next group of sync characters. The devices are then resynchronized to receive the next block of characters. The block length varies from few bytes to many hundreds of bytes.

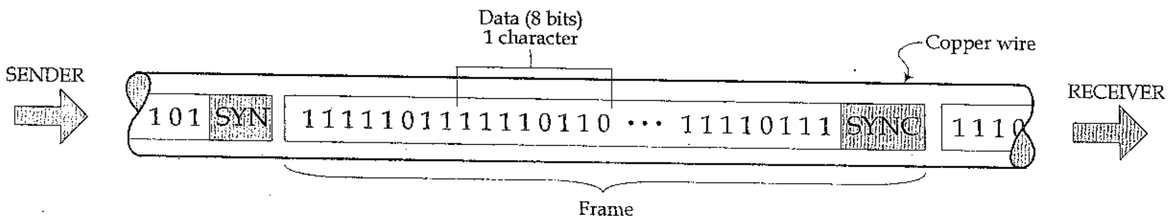


Fig.15: Synchronous Transmission

Start and stop bits for each character are not required. In other words, if you had 100 characters using an 8-bit ASCII code structure, the message part of the block of data would be 800 bits long.

Synchronous transmission is more efficient than asynchronous because there are fewer control bits in compared to the total number of bits transmitted. The synchronization takes only 16 to 32 bits (2 to 4 characters), whereas the stream of bits for the data block may be several thousand bits long. Examples of synchronous transmission are internet telephone calls, internet chat or instant messaging.

Asynchronous Characteristics

(i) Random transmission of data units. That is there is no connection between any two characters and a transmission gap may exist between two characters.

(ii) No timing (synchronization) between sender and receiver. It is triggered by the start bit and is terminated by the stop. Once a start bit is received by the receiver, it will sample the succeeding data bit to determine the binary value.

(iii) The transmission speed is lower than synchronous and is ideal for low volume data.

(iv) Data format is framed, as it is guarded by start and stop bits.

Synchronous Characteristics

(i) Data is transmitted in blocks. No gap may exist between two characters.

(ii) Synchronization must be established between the sender and receiver before the data block is sent. This is done by sending a group of sync characters to the receiver.

(iii) Transmission speed is very fast which makes it ideal for high speed and high volume data.

(iv) Data format is not framed, since it is not guarded by start or stop bits. It in blocks or packets.

3.2 TRANSMISSION MEDIA

Transmission media is used to provide a connection between sender and receiver in order to exchange information. They are grouped into two major categories namely ***Guided*** and ***Unguided***.

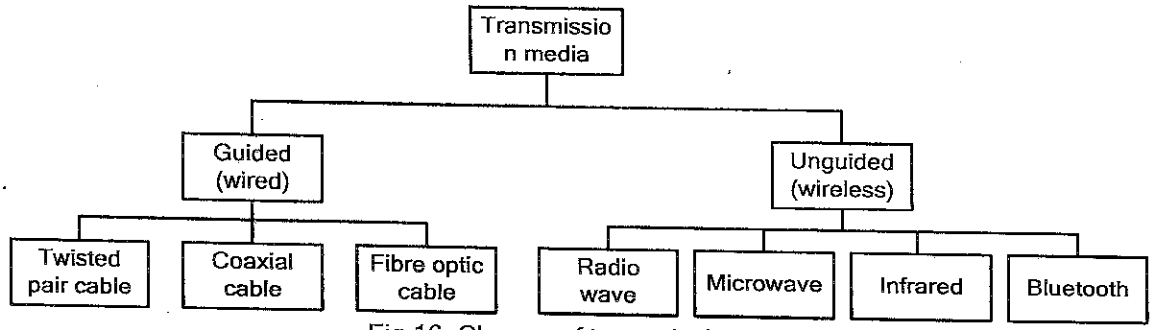


Fig.16: Classes of transmission media

Type	Description
Guided:	Data is sent through a physical and tangible connection between two communicating points. They are transmitted by means of electrical signals or light waves. Examples are twisted pair cables, optical fibre, waveguide, and coaxial cable.
Unguided:	Data is sent through space or the atmosphere. There is no direct physical connection between two points. Data is transmitted by means of electromagnetic waves and required and antenna for transmission. Examples are microwaves, radio waves, and infrared waves.

GUIDED TRANSMISSION MEDIA

(1) Twisted Wire Pair

Twisted wire pair consists of two conductors (normally copper), each with its own plastic insulation (like PVC) twisted together and packed close in a bundle. One wire is used to carry signals to the receiver, and the other is used only as a ground reference.

The wire pairs are twisted to minimize *electromagnetic interference (EMI)* and *radio frequency interference (RFI)* between one pair and any other pair in the bundle and also to reduce *crosstalk.*

Electromagnetic Interference are unwanted (noise) signals that can be generated by electric motors, fluorescent lights, medical devices and more which interfere with data transmission. The noise produced by these devices degrades data signals, sometimes stopping communication completely due to excessive error rates.

Radio Frequency Interference: The frequencies at which LANs operate fall into the range of radio signals. If twisted pair cable is not sufficiently twisted, it can function as an antenna and radiate significant amounts of radio signals that can interfere with local broadcast reception equipment causing **radio frequency interference.**

Near End Crosstalk (NEXT): The unwanted sounds or other signals picked up by one pair of wire from other pairs of wires in electronic communication systems.

Twisted-pair cabling has become a popular choice for data transmission in LANs. From technology standpoint, twisted-pair cabling consists of several two-wire pairs enclosed in synthetic sheath. Each of the twisted pairs is composed of two insulated 22- or 24-guage copper wires twisted together to reduce RFI, EMI, and crosstalk.

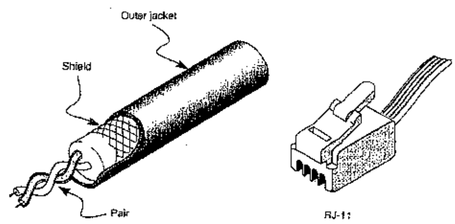
From business perspective, twisted-pair cabling is very popular because it's relatively inexpensive and easy to install and maintain. It is highly reliable, supports high data rates (up to 10Gbps for Cat 6 UTP), and it is a standard for Ethernet networks that has become common in the business world for LANs.

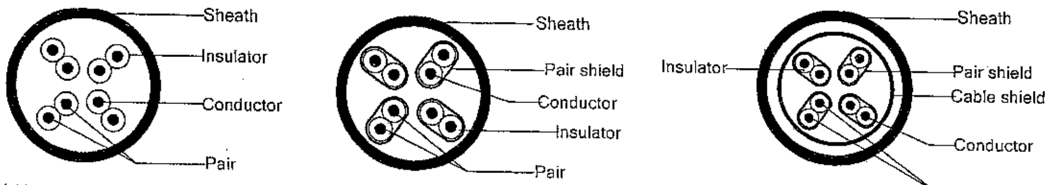
There are three types of twisted pair cables: -

(i) Unshielded Twisted Pair (UTP)

(ii) Shielded Twisted Pair (STP).

(iii) Screened Shielded Twisted Pair (S/STP)





(a) Unshielded twisted pair (UTP) *(b) Shielded twisted pair (STP)* *(c) Screened shielded twisted pair (S/STP)*

Unshielded Twisted Pair (UTP): This cable is not shielded meaning it does not have any metallic foils or braided-mesh covering for extra protection against RFI. It is used mainly for telephone and computer networking. This results in high degree of flexibility as well as rugged durability. Twisted pair cables are often used for short and medium length connections because it is cheaper than fibre and coax cables. UTPs are most commonly used in computer networking and are called "Ethernet Cables".

Shielded Twisted Pair (STP): This has a shield to prevent EMI and the shield can also serve as ground since it is a conductor even though a special grounding wire is added called a "drain wire". The purpose of a shield is also called "screening". For the shield to work it must be grounded. Both UTPs and STPs are made up of 100, 24 AWG solid conductors.

CONNECTORS: The most common connector for UTPs is called ***Eight-pin connector*** or ***RJ-45*** (RJ stands for registered jack) connector as shown in the figure above. The RJ-45 is a keyed connector, meaning the connector can be inserted in only one way. It has 4 pairs of wires. The other type, RJ-11 connector has 2 pairs of wires and it is used mainly with telephone lines.

Advantages of Twisted Pair Wiring:

(1) Telephone cable standards are mature and well established. Materials are plentiful, and a wide variety of cable installers are familiar with the installation requirements.

(2) It may be possible to use in-place telephone wiring if it is of sufficiently high quality.

(3) UTP represents the lowest cost cabling. The cost for STP is however higher and is

comparable to the cost of coaxial cable.

Disadvantages of Twisted Pair Wiring:

(1) Because of its high impedance a repeater is required every 3 km

(2) STP can be expensive and difficult to work with.

(3) Compared to fibre optic cable, all twisted pair cable are more sensitive to EMI. UTP especially may be unsuitable for use in high-EMI environments.

(4) Twisted pair cables are regarded as being less suitable for high data rates and high-speed transmissions compared coax or fiber optic cable.

(5) Cable segment lengths are also more limited with twisted pair.

(2) Coaxial Cable

When the transmission rate of the twisted pair is increased, they mainly suffer from a problem called *skin effect*, a pure electrical phenomenon. This problem greatly reduces the transmission distance (attenuation increases). Hence to operate at rates higher than 1 Gbps, a new transmission medium is needed. Coaxial cables provide a solution to this problem.

Coaxial cable (or coax) consists of a solid copper in the middle (the inner conductor) with an outer cylindrical shell for insulation, an outer shield (the second conductor) just below the protective outer jacket. Coaxial cable uses its inner conductor as one of the two conductors and the braided shield (outer conductor) as the second conductor to complete the circuit.

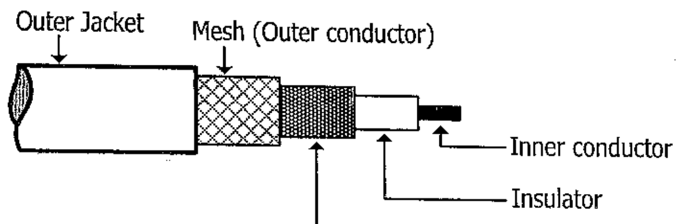


Fig.21. Coaxial Cable

Category	Impedance	Use
RG-59	75 Ω	Cable TV, Satellite-TV connection
RG-58	50 Ω	LAN (Thin Ethernet)
RG-11	50 Ω	LAN (Thick Ethernet)

CONNECTORS: To connect coaxial cable to devices, we need coaxial connectors. The most common type of connector used today is the **Bayone-Neill-Concelman (BNC)** connector. Each end of a coax has a BNC connector. Two coax cables are connected using a Barrel (BNC) connector and a BNC-T connector is used to branch out a connection to a computer or other device.

Coaxial cables are available in 52 ohms, 75 ohms, 93 ohms and a number of impedances in between. Coaxial cable usually supports frequencies up to 500 MHz. It can support very high speed for data transmission than twisted pair. Coaxial cable is used for long distance communication and TV system between the antenna and TV set. It can be used for over longer distances about 186 metres instead of 100 meters for twisted pairs.

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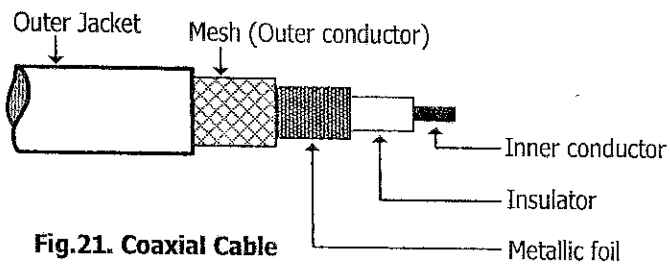

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Coax is not very flexible, so it cannot be bent around sharp corners easily. Unlike twisted pair, coaxial cable allows us to send information a greater distance before an amplifier is required. Coaxial cable is, by definition, shielded because its outer conductor is also a shield. Some coax cables are doubly shielded; an example is Ethernet whose coaxial cables generally have both a foil shield and a second braided conductor shield. The shield reduces RFIs and EMIs and also it eliminates crosstalk.

Coax cables can handle higher speeds than twisted pairs. This is because it has lower impedance than twisted pair. Most coaxial cables have an impedance of 60 to 75 ohms, whereas twisted pairs can have double that amount of impedance as high as 130 ohms. Lower impedance means the wire impedes or resists the data signal less than higher impedance. Because it encounters less resistance, coaxial cable can carry more data. It can also carry a higher bandwidth, which means it can move data at higher speeds and also can carry multiple signals simultaneously. Coaxial cables can support broadband and baseband signals.

(3) Fibre-optic Cable

Fibre-optic cable (or optical fibre) is a guided transmission medium made up of hair-thin strand of glass used to transmit data signals by means of light. It consists of very thin (hair-thin) inner core of pure glass or plastic surrounded by a cladding also of glass but having a lower refractive index than the core. The cladding is covered by an outer layer padding of Aramid wool, followed by a PVC sheath or any other insulating material to protect the core and the cladding from damage due to heat, mechanical stress and exposure. The core is designed to have a high refractive index enclosed by a cladding of lower refractive index. Light emitting diode (LED) or laser diode is used to convert electrical data signals into streams of light pulses that are injected at very high speeds into the hair-thin core of glass. At the receiving end, a photodiode detects the light pulses and reconverts them back into electrical data signals so they can be processed by conventional microcomputers/terminals. Fig.22 shows a typical cross-section and side view of fibre-optic cable showing the components layers. It is used in modern communications systems to transmit information in the form of pulses of light.

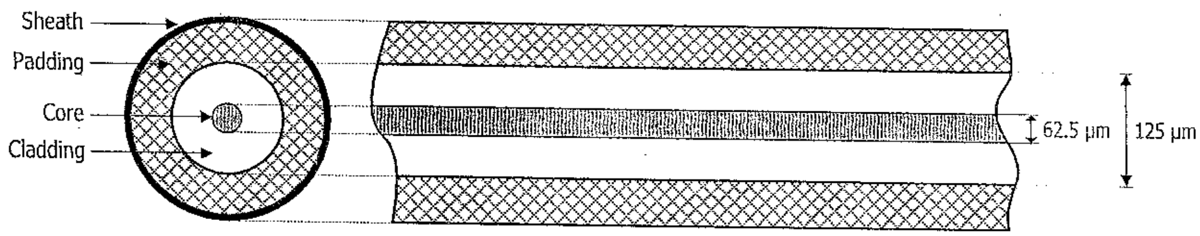


Fig. 22. Cross section of fibre-optic cable showing component layers

LEDs can generate signals up to **300 Mbps** while lasers generate signals in the **multi-Gbps** range. Hence, LEDs are used for short distance links like enterprise backbone networks while lasers diodes are used for long distance networks and also for long-haul backbone links. Optical systems operate in the **infrared frequency range** of the electromagnetic spectrum.

In fibre-optic transmission, data signals (information) are transmitted by the principle of **Total Internal Reflection** of light pulses which takes place inside the core at the core-cladding interface of the fibre-optic cable. Fig. 23 shows light pulses propagating in the core of fibre-optic cable by means of total internal reflection.

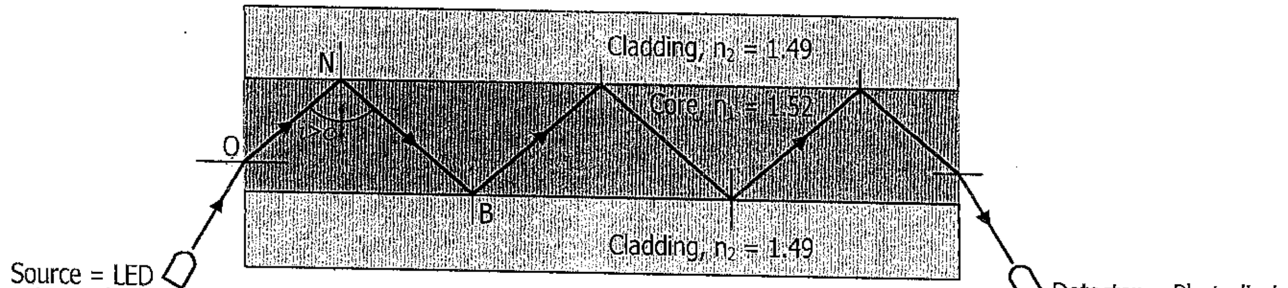


Fig. 23. Transmission of light in optical fibre by total internal reflection. It occurs when $i > c$

Types of Fibre-optic Cable

Fibre-optic cables are classified by their refractive index profile and the mode of light propagation that it can transmit. Mode refers to the path which light rays follow through the fibre depending on its angle of incidence at the core-cladding interface. The number of modes depends on the diameter of the core and can be reduced by reducing the diameter of the core. Based on this approach, there are three types shown in Fig. 24.

(1) Multimode step index (2) Multimode graded index (3) Single-mode step index

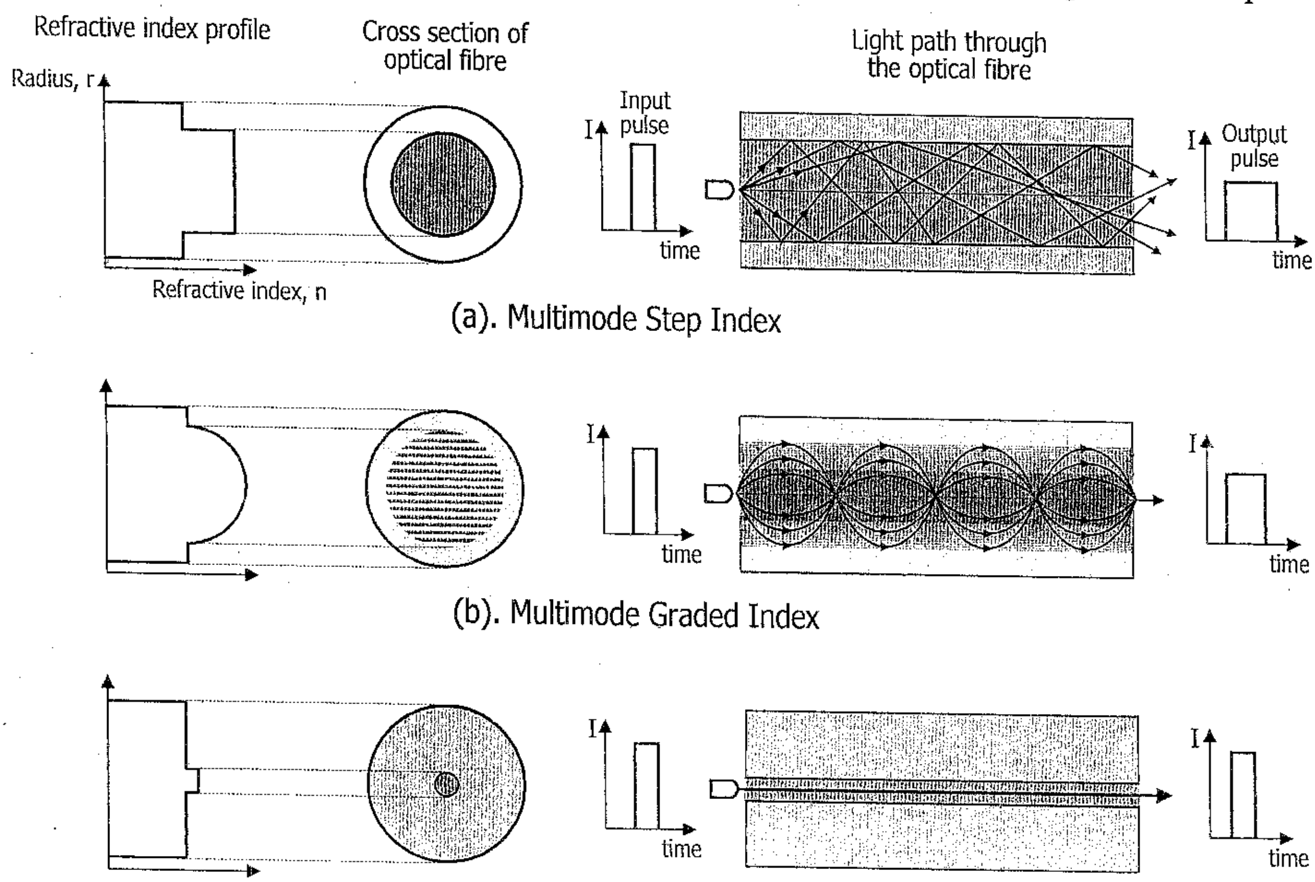


Fig. 24. Types of fibre-optic cable and the different modes of light propagation

Multimode Optical-fibres. Multimode fibres have a large core diameter of about 50 to 100 microns (micrometer, 10^{-6} m) and a total diameter of about 125 microns. It supports multiple propagation paths (modes). Each ray from the input pulse follows a different path and arrives at the receiver at different times making the output pulse to spread out in time resulting to distortion, dispersion, and attenuation of the pulse. These defects limit the transmission speed (data rate) of the cable and the maximum transmission range (cable length) before the data becomes completely distorted. This type of optical-fibre is best suited for transmission over short distances. Multimode fibres exist in two forms; multimode step index and multimode graded index.

(1) **Multimode Step Index.** This is the simplest, oldest and least expensive fibre-optic cable; it has a large core of high refractive index surrounded by a cladding of lower refractive index, with a step (abrupt) change in refractive index between the core and cladding as shown in Fig.24a. This fibre offers different multiple propagation paths, has the highest distortion, dispersion and attenuation hence, it has lowest cable length, bandwidth and, data rate. The cable length is limited to about 550 m and data rate up to 50 Mbps.

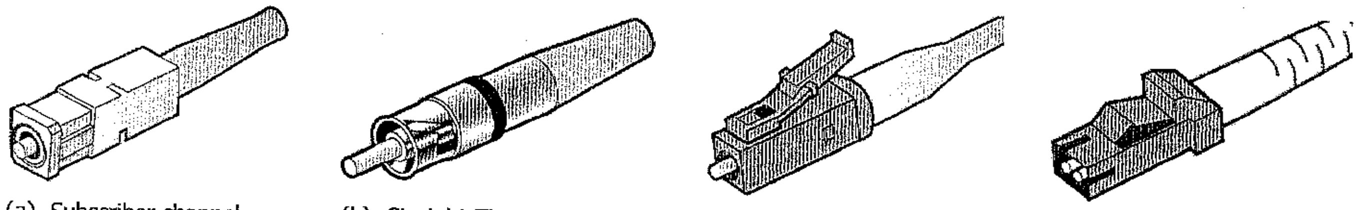
(2) **Multimode Graded Index.** This fibre has a large core with refractive index that decreases gradually with radial distance from the centre of the core to the cladding such that there is no abrupt change between the core and the cladding. The change occurs gradually and involves several layers, each with a slightly lower index of refraction. The refractive index profile is curved outwards (parabolic) from the cladding to the core. This is shown in Fig.24b. The graded index makes the rays following longer paths move slightly faster than those following shorter paths such that though the rays follow different paths, they all move at approximately the same time. They follow a wavelike path along the fibre. The net result is that the received pulse does not spread out much, therefore; the distortion, dispersion and attenuation of the output pulse is much lower than in the step index mode and the bandwidth for data transmission, and cable length are higher. It can transfer up to 1 Gbps and the cable length is up to 1,000 m. It is commonly used for LAN backbone wiring.

(3) **Single-mode (monomode) Step Index.** This has the highest performance, bandwidth and cost. The core is extremely narrow between 5 to 10 microns in diameter such that all light rays take only a single path (single-mode propagation) through it to the destination. Hence, the output pulse is exactly the same as the input pulse and distortion, dispersion and attenuation are totally minimized. The cladding is much wider than the core and there is a step reduction in refractive index between the core and cladding as shown in Fig. 24c. This fibre is the most difficult to install because it requires the greatest precision and uses injection laser diodes (ILD) as light source. However, it has the highest bandwidth and data rates, up to 1000 Gbps and higher is possible. It is typically used for long distance communications (e.g. international submarine cables) and very high-bandwidth applications. Maximum transmission range before a repeater is required is up to 100 km.

CONNECTORS

Connectors used with fibre-optic cables depend on the light source used to generate light pulses and the corresponding light detector. Three types of fibre-optic cable connectors are:—

(a) **Subscriber Channel (SC) Connector.** This uses a push-pull locking mechanism to make a strong connection. It is easy to install and requires less space for an attachment. It is used for connecting cable TV.



(a). Subscriber channel (b). Straight-Tip (c). Locking Connection (d). MT-RJ connector

Fig. 25. Fibre-optic cable connectors

(b) **Straight-tip (ST) Connector**. Uses a half-twist bayonet type of lock that locks onto the jack when twisted. They are used in Ethernet networks for connecting fibre-optic cable to networking devices.

(c) **Locking Connection (LC) Connector**. Has a flange on top similar to an RJ-45 connector for latching to secure a connection. They are about half the size of SC connectors which makes them good for high-density networking applications, where many fibres are concentrated in one location.

(d) **Mechanical Transfer Registered Jack (MT-RJ).** This looks slightly like an RJ-45 connector. It is a popular connector for terminating two fibre cables. Compared to other connector types, MT-RJ connectors take only half the space for the same number of cable terminations.

Fibre-optic Specification

Fibre-optic cables are specified in terms of their core and cladding diameters. This is usually done using fractional numbers to specify the core and cladding diameter. For example, a 62.5/125 cable has a core diameter of 62.5 microns (micrometer, 10^{-6} m) and the cladding diameter is 125 microns. The following are some commonly used fibre-optic cable configurations:

Fibre-optic cable sizes

Configuration	Core (microns)	Cladding (microns)	Mode
8/125	8	125	Single-mode, step index
50/125	50	125	Multimode, step index
62.5/125	62.5	125	Multimode, graded index

Long haul cables are bundled with about 100 - 800 fibres/cable. **NOTE**: to join fibre-optic cables, the two cladding diameter sections must be the same. Any misalignment in fibre connections will cause energy to be lost as light crosses the joint.

Advantages of Optical Fibre

(1) **Higher bandwidth**: Fibre-optic cable can support higher bandwidth (high data rates) than twisted-pair or coaxial cable. Data rates and bandwidth utilization over fibre-optic cable are limited not by the medium but by the signal generation and reception technology available.

(2) **Less signal attenuation**: Fibre-optic transmission distance is significantly greater than that of other guided media. A signal can run for 50 km without requiring regeneration. We need

repeaters every 1 km for coaxial and 3 km for twisted-pair cable.

(3) **Immunity to electromagnetic interference**: Electromagnetic noise cannot affect fibre-optic cables.

(4) **Resistance to corrosion** Glass is more resistant to corrosive materials than copper.

(5) **Light weight**: Fibre-optic cables are much lighter than copper cables.

(6) **Greater immunity to tapping**: Fibre-optic cables are more immune to tapping than copper cables. Copper cables create antenna effects that can easily be tapped.

Disadvantages

(1) Installation and maintenance: Fibre-optic cable is a relatively new technology. Its installation and maintenance require expertise that is not yet available everywhere.

(2) Unidirectional light propagation: Propagation of light is unidirectional. If we need bidirectional communication, two fibres are needed.

(3) Cost: The cable and interfaces are relatively more expensive than those of other guided media. If the demand for bandwidth is not high, often the use of optical fibre cannot be justified.

UNGUIDED TRANSMISSION MEDIA

Unguided media carry data signals in the form of electromagnetic waves that propagate through the space or air without using a physical conductor. They are not guided or confined to a channel to flow. This type of communication is referred to as **wireless** communication. The term **"wireless"** means **radio transmission.** When a laptop connects into the network wirelessly, it is using radio transmission. Signals are normally broadcast through free space at a specific frequency and thus are available to anyone who has a device capable of receiving them.

Radio data transmission uses the same basic principles as standard radio transmission. Each device or computer on the network has a radio receiver/transmitter that uses a specific frequency range that does not interfere with commercial radio stations. The transmitters are very low power, designed to transmit a signal only a short distance, and are often built into portable computers or handheld devices such as phones and personal digital assistants.

Radio Wave Propagation

Radio wave propagation is a system of communication employing electromagnetic waves propagated through space at a specific frequency and are therefore available to anyone who has a device capable of receiving them. Because of their varying characteristics, radio waves of different wavelengths are employed for different purposes and are usually identified by their frequency. The shortest waves have the highest frequency; the longest waves have the lowest frequency. In a vacuum, all electromagnetic waves travel at a uniform speed of about 3×10^8 m/s.

A radio wave can be represented in terms of *frequency* (f) and *wavelength* (λ) and *velocity* (v). If the radio wave propagates at a velocity of c (m/s) through space then the relationship between c, f, and λ

is given as: $c = f.\lambda$

Radio waves can be classified as shown in table below:

Frequency Range	Designation	Abbr.	Wavelength Range	Uses
3 - 30 kHz	Very Low Frequency	VLF	100,000 - 10,000 m	Long distance point-to-point communication,
30 - 300 kHz	Low Frequency	LF	10,000 – 1000 m	Navigational aids, Radio beacons
300 - 3000 kHz	Medium Frequency	MF	1,000 – 100 m	AM-radio, Maritime radio, Coast guard communication
3 - 30 MHz	High Frequency (SW)	HF	100 - 10 m	Telephone, SW-radio, Facsimile, communication of all types
30 - 300 MHz	Very High Frequency	VHF	10 - 1 m	VHF TV-broadcast, FM-radio, Air Traffic Control, Radar.
300 - 3000 MHz	Ultra High Frequency	UHF	1 m - 10 cm	UHF TV-broadcast, Radar communication, satellite communication
3 - 30 GHz	Super High Frequency	SHF	10 cm - 1 cm	Microwave link, radar, mobile communication, bluetooth, Wifi, Wimax
30 - 300 GHz	Extremely High Freq	EHF	1 cm - 10 mm	Railway service, Radar landing system

Properties of Radio Waves

(i) The longer the wavelength the further it goes

(ii) The longer the wavelength, the better it travels through and around things

(iii) Shorter wavelengths (high frequencies) carry more data

(iv) At low frequencies (less than 1 GHz) it is omni-directional (e.g. broadcast radio).

(v) At high frequencies (greater than 1 GHz) they are beamed or unidirectional (point-to-point).

(vi) Attenuation

(vii) Reflection

(viii) Diffraction

(ix) Interference

(x) Polarization

(xi) Absorption

Propagation Modes

Radio frequency wave travel along three paths:

(1) Ground wave

(2) Sky wave

(3) Line of Sight

Ground wave

- Travels along the curvature of the earth.
- Have carrier frequencies up to 3 MHz
- Travels very far because they are diffracted as they travel
- Does not support high data rates
- Uses tall antennas and high power transmitters
- Used by AM radio station

Sky wave

- Travels by means of reflection and refraction process in the ionosphere.
- Have carrier frequencies between 3 - 30 MHz.
- Covers long distances as signals are refracted by the ionosphere and sent back to earth.
- Does not support high data rates
- Example BBC World service, Voice of America (VOA)

Line of Sight (L-O-S)

- Travels through line-of-site process.
- Have carrier frequencies above 30 MHz.
- Distance covered is affected by obstructions and earth's curvature.
- Used by GSM phones, terrestrial microwave, satellite.

Microwave Propagation

These are short wavelength, very high frequency radio signals occupying the 1 GHz to 1 THz range and transmitted over a direct line-of-sight path between two points. Because they have very short wave lengths they can carry more data.

Characteristics of Microwave Transmission

(i) Because microwaves are electromagnetic waves they exhibit some similar characteristics e.g. Attenuation, Reflection, Diffraction, Interference, Polarization, Absorption.

(ii) Travel through L-O-S process

(iii) Can be affected by obstructions and earth curvature

(iv) Microwave repeater stations are placed approximately 40 - 80 km apart for low frequency wavelengths (10 to 100 cm wavelengths).

(v) Short wavelengths, high frequency microwaves (1 to 5 cm wavelengths, greater than 10 GHz) are limited to less than 8 km.

Application of Microwave Transmission

- **Mobile phones** operate in the 900 MHz, 1800 MHz and 1900 MHz range.
- A **Wireless LAN** operates in the 2.4 GHz range (for 802.11 b/g radio standards) and 5 GHz (802.11a radio standard) data rates of up to 54 Mbps.
- **Terrestrial Microwave** earth-based microwave systems that uses parabolic antennas to transmit and receive radio signals in a line-of-sight path between the antennas spaced at approximately 48 km apart. The antennas are usually placed on top of buildings, towers and hills to avoid obstruction.

Satellite Microwave

Use the range 1.6 GHz - 30.5 GHz for uplink and downlink. The use of Satellite is to extend the coverage area. Signal is transmitted up and down between ground stations. The satellite is therefore used as a repeater for re-generating the signal. Figure 40 shows how it works. Here, a transmit signal is reflected by the satellite to cover a region on the earth.

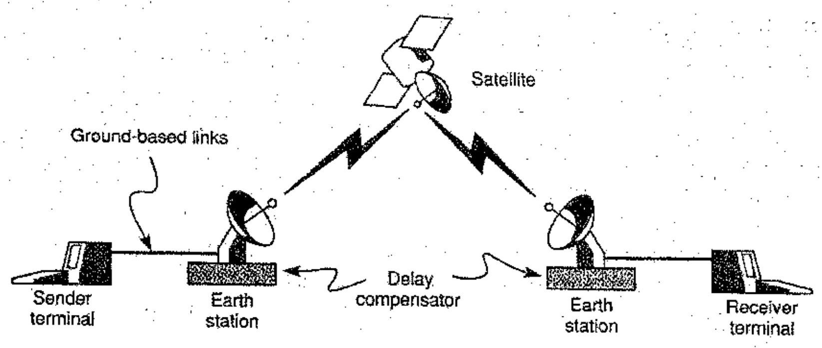


Fig.40: Satellite link

Some characteristics of satellite communications are:

(i) LOS is required.

(ii) It uses bandwidth between 4-6 GHz as C-band, 12-14 GHz as Ku-band and also the 20-30 GHz as Ka-band.

(iii) Signal requires amplification due to attenuation after travelling from the ground station to the satellite and vice versa.

(iv) Similar to microwave, the transmission quality is also subject to weather changes.

Infrared Wave

Infrared wave falls between radio waves and visible light with frequencies from 300GHz to 400THz, used for short-range communications. It can't be seen but generates heat. They are commonly used in remote controls but when used in networking, it may either be direct PTP or diffused (many-to-many). PTP mode uses laser beams requiring clear LOS between transmitter and receiver. Diffused mode spreads the light out and bounces it off walls, building etc. so that it goes to its destination and these are secure from intruders. They don't travel far.

When compared to microwave, infrared's biggest disadvantage is that it is much more susceptible to attenuation from smoke, fog, and heat wave shimmering. It is not susceptible to attenuation caused by rain, which is the primary weather condition affecting microwave. Infrared signals are immune to EMI but suffer interference from lighting.

Both infrared and microwave are reliable for short distances (800 metres or less) but microwave works well over 16 km when weather conditions are good. Infrared has several important advantages over microwave, and these may outweigh its disadvantages.

- It needs a far smaller beam-clearance area because infrared beams do not diverge or spread out much.
- Infrared signals also can reach speeds of 45 Mbps, making them useful in many video applications.
- Devices using infrared tend to be small and relatively easy to set up.
- The greatest advantage infrared has over microwave, however, is the lack of government regulation related to its use.
-

Bluetooth

This is a short range wireless digital standard aimed at linking computers, PDAs, mobile phones and accessories up to distances of 30 ft. They transmit in the range of 57.6 kbps to 1 Mbps

Advantages of Unguided Transmission Media

(1) It is ideal for areas where cable installation is difficult and expensive, and for mobile users.

Disadvantages of Unguided Transmission Media

(1) There is bandwidth limitation for frequency use.

(2) Storms or electrical disturbances produce irregular phenomena in the propagation of radio waves.

(3) Not secure. Anyone with a receiving device can intercept the signal.

MODULE FOUR

COMMUNICATION SOFTWARE AND PROTOCOLS

Introduction

Software generally comprises all programs (instructions) that cause the hardware (machines) to do work. It enables the user to communicate (interact) with the hardware and cause the machine to perform some tasks. The hardware devices (computer, modem, multiplexer, FEP) in a computer network must be loaded with software (that is, communication software) before the devices are able to communicate with each other.

Communication Software consists of all programs that enable both user and hardware components in a computer network to connect with each other in order to send and receive data through telecommunication channels (telephone lines, microwave and fibre optics etc) using a modem. Communication software coordinates the flow of information between all interconnected components in a computer network. It is located at various points in the network and depending on where they are located they are responsible for supporting applications in moving information from one point to another in the network.

For example in figure 16, the host computer might have operating system, teleprocessing monitor software, database management systems, security software and application programs. The front end processor (FEP) has telecommunication access programs for network control. The remote intelligent terminal controller, switch, router, statistical multiplexer and concentrator may have switching software, store and forward software and control software packages that perform a subset of the telecommunication access program functions located at the FEP. At the remote end of the network, there might be software packages located at or within a microcomputer so it can access the network.

Microcomputers can accept more generalized software programs and, therefore, can also perform different specific application functions. All these pieces of hardware mentioned above use specialized communication software programs to perform specific communication tasks required for network control and data movement. The software programs located at the remote end of the network often are scaled-down subsets of the software located in the FEP or the host mainframe computer located at the central site.

Types of Communication Software

The various communication software mentioned above can be grouped into four categories: -

(1) Application Software

(2) Teleprocessing Monitor Software

(3) Telecommunication Access Software/Line Control Software

(4) Performance Software

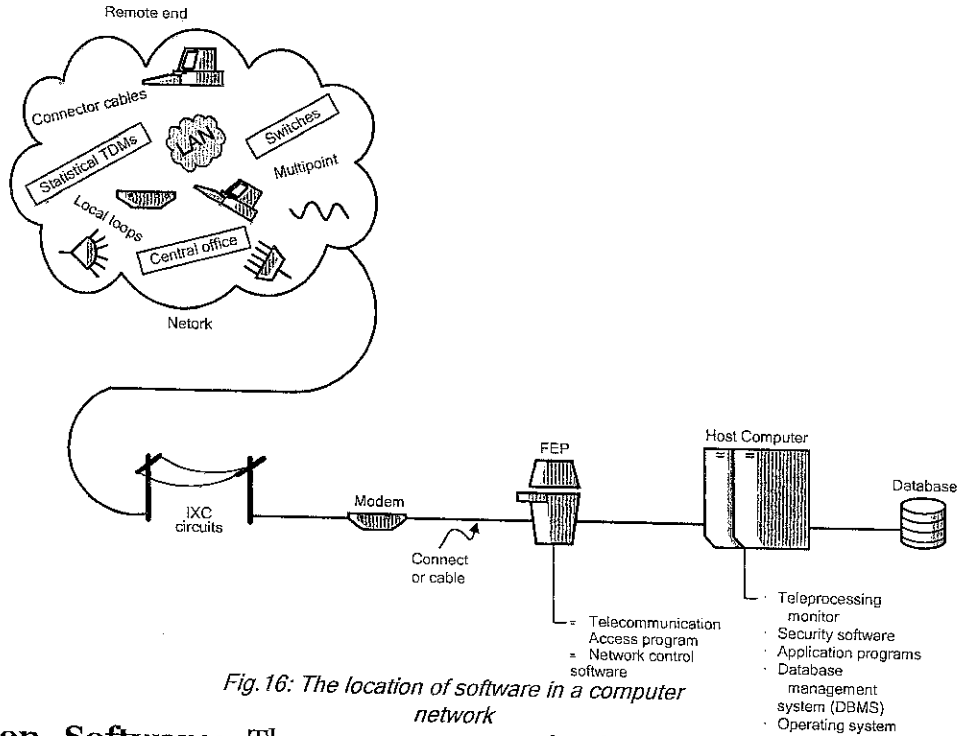


Fig. 16: The location of software in a computer network

(1) Application Software: These are communication application programs installed in a microcomputer, host or FEP. It is used to provide remote access to other systems in order to exchange messages and data files in text, audio, and/or video formats. This includes email programs, Internet browsers, terminal emulators, file transfer programs, chat and instant messaging programs.

Functions:

Provide access to the Internet or Intranet server acting like a web browser. Software packages such as Microsoft Internet Explorer and Mozilla Firefox provide users with an easy-to-use interface to the Internet. A web browser is a program that enables a computer user to find and display information and web sites on the internet. Other examples are Netscape Navigator, Google Chrome and Mosaic.

Using file transfer protocols to perform file transfer. Protocols such as File Transfer Procotol (FTP) and Hypertext Transfer Protocol (HTTP) are included with communication software applications that allow microcomputer users to transfer information from the Internet to their computer. The FTP application is automatically invoked when a user downloads a file from the Internet. When a web browser encounters a web site that begins with ftp: //, it automatically uses FTP to access the information.

Terminal Emulation. This function of communication software allows users to log on to a remote computer. A remote log in establishes a remote user as a local user on the remote computer. Once a user has successfully logged on to the remote computer, the services available on the computer can be used and the results transferred back to the local computer. When the user logs off from the remote computer his session is re-established on the local computer.

A Software package designed to simulate a user as terminal on a remote computer is called Terminal Emulator. TELNET (TErminaL NETwork) is a general-purpose communication software with very good terminal emulation capability.

Electronic mail. This feature allows the transmission of electronic messages and data between computers via a network. Examples of email software include Microsoft Outlook, Pegasus Mail, and Eudora.

(2) Teleprocessing Monitor Software: This software resides in the host computer; it monitors the transfer of data between multiple terminals and application programs to ensure that the transaction processes completely or if an error occurs, to take appropriate actions. It is frequently used for mainframe-based wide area networks, where Teleprocessing Monitors directly relieve the host computer's operating system of many tasks involved in handling message traffic between the host and the front end processor or the host and other internal CPU software packages (e.g. DBMS).

Functions

- Placing messages into input or output processing queues
- Access methods
- Task scheduling
- Error handling within the host,
- Handling restart/recovery for problems

Examples include the CICS (Customer Information Control system) for IBM mainframes. CICS can perform thousands of transactions per second. ENCINA and BEA-Tuxedo are major teleprocessing monitors in UNIX client/server system.

(3) Telecommunication Access (line control) Software. This use to reside in the host computer but now resides in the front end processors, switching nodes (SNs) of a packet network, remote intelligent terminal controllers or microcomputers. This software provides the following functions. An example is IBM's Network Control Program (NCP).

Functions:

- Polling and selection of terminals in a central control network. Polling is asking each terminal whether it has a message to send, and selecting is asking each terminal whether it is ready to receive a message.
- Automatic dial-up and answering calls
- Code conversion
- Message switching or store and forward switching
- Circuit switching and port contention
- Logging of all inbound and outbound messages
- Error detection and retransmission when an error is detected.

(4) Performance Software. Network performance software normally resides in the host computer, FEP or microcomputer. It refers to a set of software programs that can be used to gather information about the functioning of network resources. Network Performance Software may focus more on network data transmission or instead on the performance of network-enabling devices such as switches and routers. It is designed to help network administrators to monitor and manage the network. Examples of performance software are Orion's Network Performance Monitor – (Orion NPM v9.5.1) and Novell's Netware Management System.

Functions:

- Network Device Monitoring. This feature is used to monitor the real-time performance and health of the devices on the network. It is able to determine the status of network components such as modems, lines, terminals, multiplexers and so on.
- Real-time Net Flow Analyzer. This is a traffic analyzer that captures and analyzes the net flow of data in real time to show exactly what type of traffic are on the network, where that traffic is coming from, and where it is going.
- It gathers network statistics, such as line utilization information, together with capabilities for evaluating those statistics. The information produced assist network administrators in capacity planning.

COMMUNICATION PROTOCOLS

In computer networks, communication occurs between entities (hardware and software) in different systems. An entity is anything capable of sending or receiving information; it may be an application program like internet explorer, FTP program or it can be hardware like NIC or switch. But, two entities cannot simply send data to each other and expect to be understood. For communication to occur, the entities must agree on a strict set of rules or procedures known as Protocol.

Communication Protocol is a set of rules that are required to initiate and maintain data communication. A protocol defines what is communicated, how it is communicated, and when it is communicated. Simply put, a protocol is a set of rules that allow two or more endpoints (computer, printer etc) to communicate. The key elements of a protocol are **Syntax, Semantics** and **Timing.**

Syntax: This refers to the structure or format of the data (or bit stream i.e. series of 1s and 0s), meaning the order in which they are presented. For example a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, and the rest of the stream to be the message itself.

Semantics: This refers to the meaning of each section of bits. How is a particular pattern to be interpreted and what action is to be taken based on that interpretation? For example, an address of all 1s (1111111) can be interpreted as a broadcast message to all network terminals.

Timing: The timing refers to two characteristics: when data should be sent and how fast they can be sent. For example, if a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.

THE NEED FOR COMMUNICATION PROTOCOLS

Communication protocols are necessary for several reasons:

(1) Enables two or more processes in different machines to communicate with each other.

(2) They are associated with particular tasks such as data packaging or packet routing.

(3) A protocol specifies rules for setting up, carrying out, and terminating communication connections and,

(4) It specifies the format the information packet must have when travelling along this connection.

(5) Error control

(6) Source and destination addressing.

TRANSMISSION PROTOCOLS

In computer networks, protocols required for communication between entities are organized into layers with each sub layer performing a set of related functions required to communicate with the corresponding peer layer in another system.

The International Organization for Standardization has defined a 7-layer model to clarify various tasks in data communication systems. The model is called **Open System Interconnection Reference Model.** Each layer performs specific tasks that allow entities and devices in different systems to communicate with each other as if they are operating in the same system.

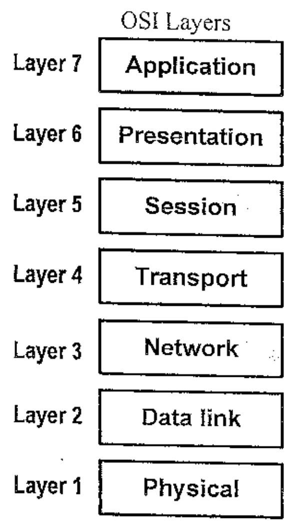


Fig. 15: OSI Reference Model

Peer layers communicate by means of formatted blocks of data that obey the set of rules or protocols assigned to the layer. Thus, referring to the OSI reference model we have; application layer protocols, presentation layer protocols, session layer protocols, transport layer protocols and so on.

TRANSMISSION PROTOCOLS are Data Link Layer protocols used to move data frames from one node to the next (i.e. node-to-node communication) until the information reaches the destination. Transmission protocol can be categorized as – **Asynchronous** and **Synchronous** transmission protocols.

Asynchronous Protocol – This is the oldest and one of the most common data-link protocols. It relies on special signals (start and stop bits) to mark the individual characters and also with its own error detection scheme, usually a parity bit. The sender and receiver are not synchronized with each other. It may also be referred to as (start-stop protocol). The frame format is shown below.

1 bit	7 bits	1 bit	1 bit
START (0)	Character	Parity	STOP (1)

Typical frame format for Asynchronous Protocol

Synchronous Protocols – It relies on timing to identify transmission characters and is most suited for transmissions that occur at a relatively constant rate. Synchronous transmission means whole blocks of data are transmitted as frames after the sender and the receiver have been synchronized.

8 bits	8 bits	8 bits	Variable	16 or 32 bits	8 bits
Beginning Flag 01111110	Address	Control	Message (Data Packet)	Frame Check Sequence	Ending Flag 01111110

Typical frame format for synchronous protocol

Beginning Frame: each frame begins and ends with a special bit pattern (01111110). This is known as the beginning and ending flag. The beginning flag indicates the position of the address and control frame elements and initiates error checking procedures.

Address field identifies one of the terminals. This 8-bit field might contain a station address, a group address for several terminals, or a broadcast address to all terminals

Control field identifies the kind of frame that is being transmitted, such as information, supervisory, or unnumbered. The information frame is used for the transfer and reception of messages. Supervisory frame is used to transmit acknowledgements, such as to indicate the next expected frame, indicate that a transmission error has been detected, stop sending and call for the retransmission of specified frames. The unnumbered frame is used for other purposes, such as to provide a command "disconnect" that allows a terminal to announce it is going down or for error correction.

Message field is of variable length and it represents the actual message.

Frame Check Sequence provides a 16-bit or 32-bit cyclical redundancy checking (CRC) calculation for error detection and correction.

Synchronous data link protocols can be divided into two groups; **byte-oriented** and **bit-oriented**. Byte-oriented protocols use predefined set of character codes to convey the information and a special character set is used for the data formatting and supervising the transfer of information across the link (link control). The most common character-oriented codes used are ASCII and EBCDIC. A disadvantage of this method is that the characters used for the link control cannot be used as ordinary data bytes. An example of character-oriented synchronous protocol is binary synchronous communication protocol.

Binary Synchronous Communication Protocol (BSC) – This is one of the oldest communication protocols developed by IBM in the early 1960s for connecting terminals to mainframes. It supports ASCII and EBCDIC. It uses special characters to indicate where frames start and end. The beginning of a frame is denoted by sending a special SYN (synchronization) character.

8 - 16 bits	8 bits		8 bits	Variable	8 bits	16 bits
SYN	SOH	HEADER	STX	TEXT	ETX	CRC

Typical frame format for BSC

The data portion of the frame is then contained between two more special characters: STX (start of text) and ETX (end of text). The SOH (start of header) field serves much the same purpose as the STX field. The frame format also includes a field labeled CRC (cyclic redundancy check), which is used to detect transmission errors.

Bit-Oriented Protocols. The data section of a frame at the data link layer might come from a sequence of bits to be interpreted by the upper layers as text, audio or video. **Bit-oriented protocols transmit individual bits without regard to their interpretation.** Such protocols can establish timing and manage data links using bit signals. Individual bits are used for timing (so that sender and receiver stay in synchrony) and also for link control. Bit-oriented protocol is more efficient than byte-oriented protocol. Synchronous Data Link Control (SDLC) protocol developed by IBM is an example of a bit-oriented protocol; SDLC was later standardized by the ISO as the High-Level Data Link Control (HDLC) protocol.

High-Level Data Link Control (HDLC) – HDLC uses essentially the same frame structure as SDLC as shown below.

8 bits	8 bits	8 or 16 bits	Variable	16 or 32 bits	8 bits
Flag 01111110	Address	Control	Message (Data Packet)	FCS	Flag 01111110

Typical frame format for SDLC and HLDC Protocol

HDLC uses a special 8-bit pattern flag 01111110 to separate one frame from the other. This sequence is also transmitted at any time that the link is idle so that the sender and receiver can keep their clocks synchronized. This pattern can create problems if the flag pattern appears in the data. We need to inform the receiver that this is not the end of the frame. This is solved by using a strategy called **bit stuffing.** In bit stuffing, if a '0' and five consecutive '1' bits are encountered, an extra 0 is added. This extra stuffed bit is eventually removed from the data by the receiver. This guarantees that the flag field does not accidentally appear in the frame.

Transmission Control Protocol/Internet Protocol (TCP/IP) Suite

TCP/IP is a group of several networking protocols developed for use on the Internet. It is a set of rules that enable different types of computers and networks on the Internet to communicate with one another.

TCP defines how data are transferred across the Internet to their destination. IP defines how data are divided into chunks, called packets, for transmission; it also determines the path each packet takes between computers.

It is an efficient file transfer protocol that permits sending of large files over sometimes unreliable networks with assurance they will not be corrupted while being transmitted.

TCP/IP is actually the oldest networking standard, developed for Advanced Research Project Agency Network (ARPANET) for computers using the UNIX operating system, but it is now used by every computer, regardless of operating system, on the Internet.

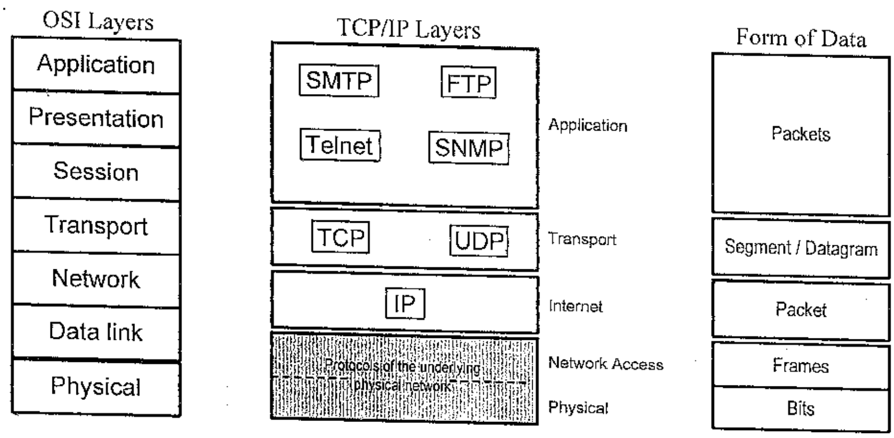


Fig. 20: Comparison of OSI and TCP/IP

The main protocols in the suite and their functions are given below:

(1) SMTP (Simple Mail Transfer Protocol) provides a simple electronic-mail (e-mail) service. SMTP uses the TCP protocol to send and receive messages.

(2) FTP (File Transfer Protocol) enables users to transfer files from one machine to another. FTP also uses the services of the TCP protocol at the transport layer to move the files.

(3) Telnet provides terminal-emulation capabilities and allows users to log in to a remote network from their computers.

(4) SNMP (Simple Network Management Protocol) is used to control network management services and to transfer management-related data.

(5) TCP (Transmission Control Protocol) provides connection- and stream oriented, transport-layer services. TCP uses the IP to deliver its packets.

(6) UDP (User Datagram Protocol) provides connectionless transport layer service. UDP also uses the IP to deliver its packets.

(7) IP (Internet Protocol) provides routing and connectionless delivery services at the network layer. The IP uses packet switching and makes a best effort to deliver its packets.

NETWORK ARCHITECTURE

In computer networks, users connected to one network often want to communicate with users attached to a different one. This means that different and often incompatible entities (hardware and software) in different systems are connected in order to exchange information. An entity is any device capable of sending and receiving information examples of entities are email programs, Internet browsers, file transfer programs, chat and instant messaging programs while a system is any device that contains one or more entities. Examples of systems include computers, terminals, and remote sensors.

An orderly exchange of information would require that the entities in each system obey some pre-established agreement or rules. These rules specify the format, relative timing of the information to be exchanged and control information for coordination between the systems. The set of agreements or rules is referred to as Protocol. A set of layers of protocols used to design and build a computer network aimed at allowing different systems, devices and software to communicate with one another is called **Network Architecture**.

In order to simplify the task of building a network that would connect different entities in different systems, the process of moving information from one node to another is planned as a stack of layers or levels with well-defined interfaces and with each one built upon the one below it and designed to perform some specific functions in the process. This is the concept of **Layered Approach** in network design.

The number of layers, the name of each layer, the contents of each layer, and the function of each layer differs from network to network. A layer refers to an entity or process or device in the network designed to provide specific services (tasks) to the layer above it while shielding those layers from the details of how the services are actually implemented.

In a layering approach as shown in fig. 15, within a single machine, the role of layer N is to offer a set of well-defined services to layer N + 1 (i.e. the next higher layer) and uses the services provided by Layer N – 1.

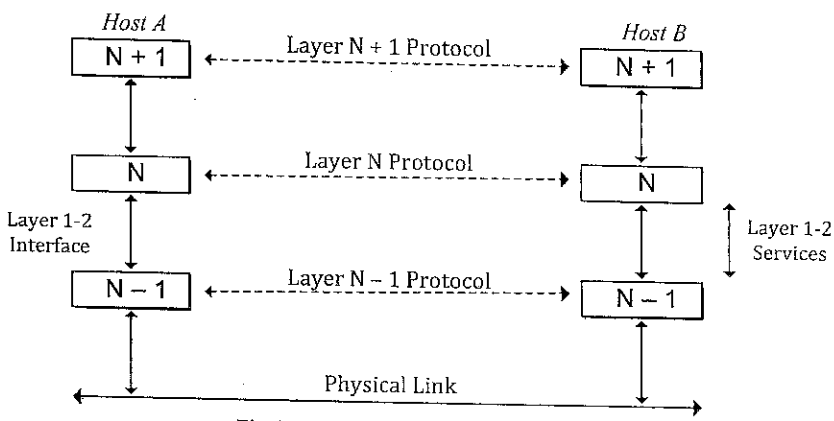


Fig.15: Layered Architecture

Layer N on one machine (or at one node in the network) communicates **peer-to-peer** with layer N on another machine. This communication is managed by an agreed-upon series of rules and conventions called **Layer-N protocol**.

NEED FOR LAYERED APPROACH IN NETWORK DESIGN

1. To simplify the complexity of network design, operation and maintenance into manageable tasks. Rather than using a single piece of software that carries out all the tasks in the process, several layers are used, each of which solves one part of the process.
2. Different implementations are able to produce compatible and interoperable networking equipment according to the layering protocols.
3. It provides a more modular design. To add some new service, only the functions of one layer need to be modified reusing the services (functions) provided at all the other layers.

Two types of network architecture styles exist:

(i) Proprietary architecture

(ii) Open architecture

Proprietary architecture refers to a design that is controlled or owned by a single vendor.

Example of Proprietary Network Architectures

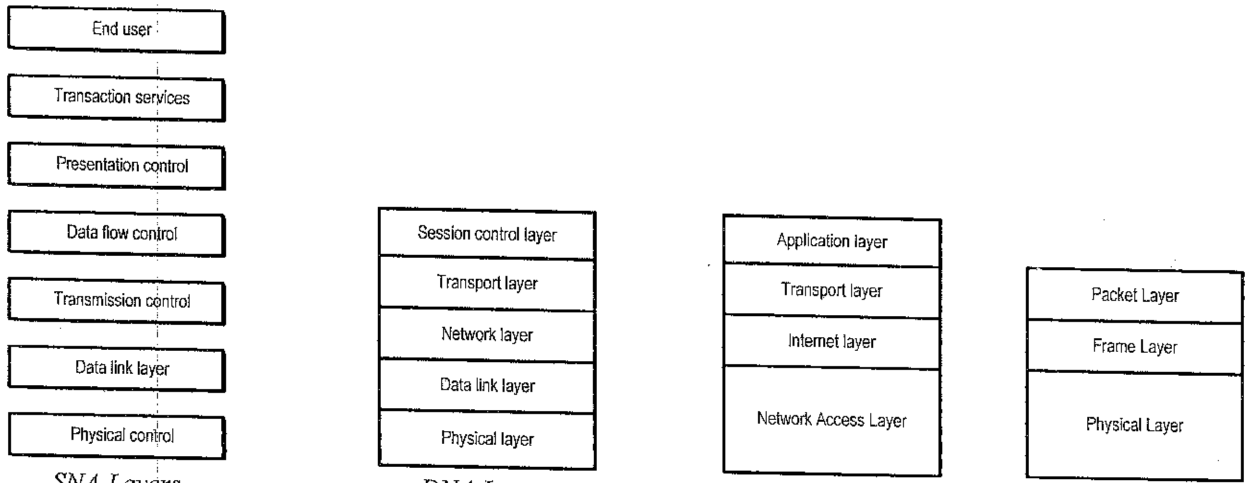
(1) Systems Network Architecture (SNA) developed by IBM.

(2) Digital Network Architecture (DNA) developed by Digital Equipment Corporation.

(3) Transmission Control Protocol/Internet Protocol (TCP/IP) model developed by ARPANET (Advanced Research Project Agency Network) in the early 1970s and sponsored by U.S. Defence Advanced Research Projects Agency (DARPA).

(4) Joint Academic Network (JANET). An electronic-mail (e-mail) network run by universities and other academic institutions in Great Britain.

(5) Attached Resource Computer Network (ARCNET) developed by Datapoint Corporation in the late 1970s.



SNA Layers *DNA Layers* *ARPANET* *JANET Layers*

An **Open Architecture** is one whose set of protocols are available to any vendor. This allows any vendor to design hardware and software components that can easily communicate with each other regardless of their underlying design. A typical example is Open Systems Interconnection (OSI) Reference Model defined by International Organization for Standardization (ISO) in 1974. This is NOT a protocol; it is a reference model for understanding and designing a network architecture that is Flexible, Robust and Interoperable.

OSI Reference Model (OSIRM)

This is a layered framework for the design of network systems that allows communication between all types of computer systems. It consists of 7 separate but related layers each of which defines a part of the process of moving information across a network.

Each layer performs specific functions which allow application software on different system architecture to communicate with each other as if they were operating on the same system. In the layered approach it is possible to use different application programs.

Layer 7	Application layer
Layer 6	Presentation layer
Layer 5	Session layer
Layer 4	Transport layer
Layer 3	Network layer
Layer 2	Data link layer
Layer 1	Physical layer

OSI Reference Model Layers

Main Objectives of the OSIRM

1. To describe how data is sent and received over a network.
2. To allow manufacturers of different systems to interconnect their equipment through standard interfaces.
3. To allow software and hardware to integrate well and be portable on different systems.
4. To create a model which all countries of the world use.

Basic Functions OSI Layers

Physical Layer: Responsible for the transmission of binary data. Provides physical connection between the computer and network.

Data Link: Physical addressing and media access. Ensures that the transmitted bits are received in a reliable way, such as adding extra bits to define the start and end of the data frame, identifying and addressing the physical address (MAC address) of Network Interface Cards (NIC), error detection/correction etc.

Network: Logical addressing and determining the best path/routing. Determines how packets are routed from source to destination network.

Transport: Also known as **"end-to-end layer"**. This layer is responsible for process-to-process transmission of data. Computers run several processes (a process is an application program running in a host) at the same time. For this reason process-to-process transmission of data means delivery from the specific application program running on one computer to the other.

Session: This layer is provides host-to-host communication. It establishes, maintains and terminates network connections.

Presentation: This layer performs data representation, compression and encryption. This layer defines the format the data uses as it is transmitted. It formats the data for the user so that it is readable and the message can be understood.

Application: It provides access to network services. Enables the user whether human or software to access the network. It is the software that the end-user interacts with. This is an application like Firefox, Outlook, or Internet Explorer.

MODULE FIVE
COMPUTER NETWORKS AND TOPOLOGY

INTRODUCTION

Network commonly means a system of interconnected lines or links. In telecommunications, network refers to a group of interconnected devices communicating with each other. The devices could be telephone exchanges, radio stations, or TV stations. These devices are interconnected by communication channels or lines, thereby forming telephone, radio and TV networks.

Therefore, **Computer Network** can be defined as a collection of computer systems and devices connected together by communications links for the purpose of exchanging information and sharing resources. The resources could be data, software or hardware. Network users are able to share files, printers, software and other resources; send electronic messages; and run programs on other computers. The devices in computer network are called **Nodes** which is any entity or device with physical or MAC address and is capable of sending and receiving data.

COMPONENTS OF A COMPUTER NETWORK

A computer network has four layers of components these are,

(1) Network hardware

(2) Transmission media (communication channels)

(3) Network Protocols and Software

(4) Network Architecture

Network hardware is made up of the physical components or devices that form the network. It consists of the computing devices (nodes) and communication electronic devices for routing and switching data from source to destination. Hardware equipments used to build computer networks include the following: -

(i). Servers

(ii). Workstations

(iii). Network Interface Cards (NIC)

(iv). Switches

(v). Hubs

(vi). Routers

(vii). Bridges

(viii). Gateways

(ix). Modems

(x). CSU/DSUs,

(xi). Multiplexers

(xii). Concentrators

Transmission Media refers to communication links that interconnect the various hardware components in the network. They are used to transmit data and control signals from one point to another in the network. They are twisted pair wires, coaxial cables, optical fibre cables, cellular phones, microwave links, satellite links, infrared and so on.

Network Protocols and Software. Protocols provide the common set of rules and procedures that defines how the exchange of information should take place. It defines the rules for initiating and terminating data transfers, interpreting how data is represented and transmitted and how errors are handled. Protocols make logical connections between network applications, direct the movement of packets through the physical network, and minimize the possibility of collisions between packets sent at the same time.

Network software consists of computer programs (set of instructions) that carry out the protocols or rules for computers to talk to one another. Examples of network software include Network Operating Systems, Client/server, and Peer-to-Peer programs. *Client computers* send requests to use network resources to other computers, called *servers* that control data and applications. In a *peer-to-peer* network, computers send messages and requests directly to one another without a server intermediary.

Other forms of network software include application programs that interface with network users and permit the exchange of information (in text, graphics, video format) between two locations for example web browsers, terminal emulators, file transfer protocols, streaming media, instant messaging programs etc.

Network Architecture is the way in which the transmission media, hardware, protocols and software are integrated to form a network by isolating the user and the application programs from the details of the network. Network architecture is the plan that connects different types of protocols, software, and networks to form a usable network. It ensures interoperability between various devices and equipment made by different manufacturers. Example IBM's Systems Network Architecture (SNA), Digital Equipment's Digital Network Architecture (DNA),

NEED FOR COMPUTER NETWORK

Computer networks can be used for several purposes:

(i) ***Facilitating communications:*** Using a network, people can communicate efficiently and easily via e-mail, instant messaging, chat rooms, telephone, video telephone calls, and video conferencing.

(ii) ***Sharing hardware:*** In a networked environment, each computer on a network can access and use hardware on the network. Suppose several personal computers on a network each require the use of a laser printer. If the personal computers and a laser printer are connected to a network, each user can then access the laser printer on the network, as they need it.

(iii) ***Sharing files, data, and information:*** In a network environment, any authorized user can easily access data and information stored on other computers on the network. The capability of providing access to data and information on shared storage devices is an important feature of many networks.

(iv) ***Sharing software:*** Users connected to a network can access application programs (such as word processors and spreadsheets) on the network.

(v) ***Centralized location of data:*** files and programs are stored on central hard disk. When a network user accesses data, it is retrieved from this central storage location. For example members of accounting department can share access to central files. They can work on records individually and then update the central files on the network.

CLASSIFICATION OF COMPUTER NETWORKS

Computer networks can be classified into several categories. Different criteria are used to classify computer networks. The following are the criteria frequently used.

(1) Technological arrangement
- (i) Client/server
- (ii) Peer-to-Peer

(2) Administrative arrangement
- (i) Centralized systems
- (ii) Distributed systems
- (iii) Collaborative systems

(3) Geographical arrangement
- (i) Local Area Network (LAN)
- (ii) Backbone Network (BN)
- (iii) Metropolitan Area Network (MAN)
- (iv) Wide Area Network (WAN)

(4) Topological Arrangement
- (i) Bus network
- (ii) Star network
- (iii) Ring network
- (iv) Mesh

Technological Arrangement:

(i) Client/server Network: This describes a computer network in which some processing is performed at the client (user's computer) and some at the server (central computer). All clients access files stored on this central computer (the server). Every client in the network can access the files stored on the server such as accessing a database.

A ***server*** is a powerful central computer connected to the network that has specialized network operating system installed and configured to provide shared network resources to users and devices on the network. The server allows users to perform functions such as store and retrieve files from the hard disk, gain access to shared printers and run shared network applications.

A ***client*** is a microcomputer connected to the network that has a local operating system installed along with special software known as ***network operating system client software*** that provides the client local operating system with additional functions required to connect to the network server to make use of the shared network resources.

In a client/server network, all the files are stored in the server. This makes the files easy to manage, back up and protect. However, if the server malfunctions, the entire network is affected.

A client/server network provides a highly efficient way to connect ten or more computers exchanging large amounts of information. Popular programs that provide client/server networking capabilities include NetWare and Windows NT.

(ii) Peer-to-Peer Network: In a peer-to-peer (P2P) network, each computer is configured to function as both a server and a client (user). It gives every other computer in the network access to all the network's resources. In a network of peers, any computer can share its hard disk and printer with any other computer without going through an intermediary like a central computer server.

All users on a peer-to-peer network store their files on their own computers. Any user on the network can access files stored on any other computer. Thus files are stored in many different locations. This makes the files difficult to manage, back-up and protect. However, if one computer malfunctions, the rest of the network will not be affected.

A peer-to-peer network provides a simple and inexpensive way to connect fewer than ten computers. Popular programs that provide peer-to-peer networking capabilities include all Windows OS, Windows for Workgroups and OS/2 Warp Connect.

Administrative Arrangement:

(i) Centralized Systems: This is server-based network with thin clients and it was built upon the foundation of mainframe and minicomputers. It is a centralized processing system in which all of the computing functions such as running applications and processing data takes place in the mainframe computer (the server). Thin clients are computers (terminals) without a central processing unit (CPU) connected to the network server and they make use of the server to run applications, process data and store data.

(ii) Distributed Systems: This is a distributed processing network, in which a task is performed by separate computers that are linked through a communications network. Instead of one single large machine being responsible for all aspect of the process, separate computers handle a subset.

This system is made up of clients and servers connected so that any system can communicate with any other system. Distributed systems are based in the enterprise network that links workgroups, departments, branches, and divisions of an organization. Data is not located in one server but in many servers and the servers can be at different geographical locations connected by Wide Area Network link.

(iii) Collaborative Systems: This system of arrangement allows users to work together on projects usually in real time. Some examples are:

- Application suites like MS Office that allows messaging, scheduling, document co-authoring, discussion groups etc.
- Video conferencing
- Internet collaboration tools that allow virtual meetings, group discussions, chart-rooms etc.
- Instant messaging.
- Workflow management involving coordination of document flow within an organization.

Geographical Arrangement

(i) Local Area Network (LAN): Local area network is a group of interconnected computers and other communication devices that are located in the same general area and connected by a common communication media so that they can share data, applications, and resources such as printer, fax machine, modem, etc. All network resources and their management activities are controlled by means of special system software called Network Operating System (NOS). The physical connection between LAN devices can be a coaxial cable, pairs of copper wires, or optical fibre. Wireless connections also can be made using infrared or radio-frequency transmissions.

For example a LAN can be used within a building, on a business premises, or in a college campus. Computers in a LAN are separated by distances of up to a few kilometres. Local area networks are differentiated by the fact that their transmission media (circuits) generally do not cross public road. Figure 24, shows a LAN located within the records building at McClellan Air Force Base in Sacramento. This eight-station LAN is used by personnel who maintain the military records within this building.

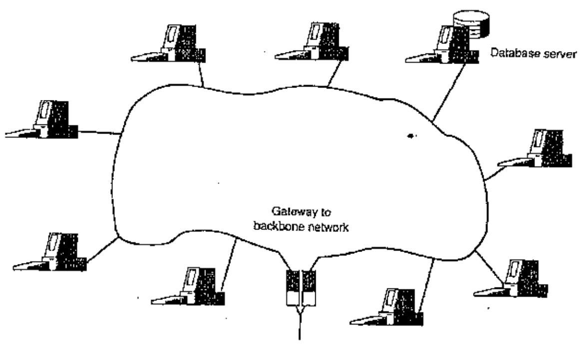


Fig.24: *Local area network (LAN) configuration. This figure shows how the local area network within the Records Building at McClellan Air Force Base connects via a gateway to the base's backbone network in fig.24*

(ii) Backbone Network (BN): Backbone network is a large central network to which all the computers within an organization are connected. Backbone networks usually connect everything on a single company or government site.

Looking at figure 24, you can see that a backbone network at McClellan Air Force Base connects six different buildings or work areas on the base. Backbone networks are similar to metropolitan area networks, except that they connect facilities covering a smaller geographical area and they normally are controlled by a single organization.

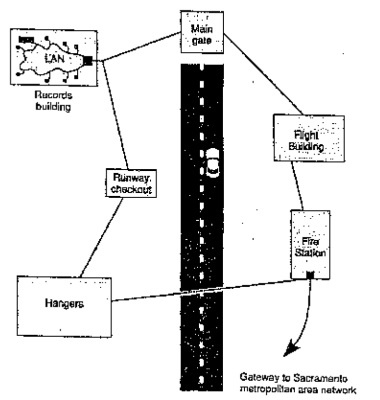


Fig.24: Backbone network configuration of McClellan Air Force Base in Sacramento. Notice the gateway to MAN that is shown in fig.25

(iii) Metropolitan Area Network (MAN): A metropolitan area network is a network larger than BN. It spans an area that covers a city or country. The various nodes may be anywhere from a few tens to hundred kilometres.

MANs interconnect various buildings or other facilities within a citywide area. Different hardware and transmission media are often used in a MAN for efficient transmission of information. In the example shown in figure 25, this MAN is within the city of Sacramento.

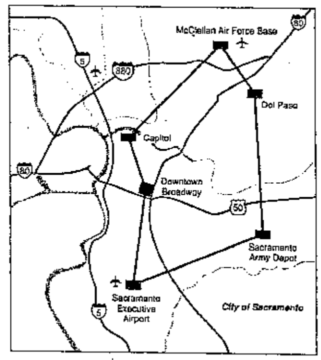


Fig.25: Metropolitan area network (MAN). This figure shows how McClellan Air Force Base interconnects with five other nodes in Sacramento.

(iv) Wide Area Network (WAN): This is a computer network covering a wide geographic area such as a city, state, country, or spans even national boundaries using a communications channel that combines many types of media such as telephone lines, cables and air waves. A WAN often uses transmission facilities provided by common carriers, such as telephone companies. It uses dedicated, switched (dial-up) and broadband connections to link computers at remote locations.

WANs are implemented to connect large number of LANs, BNs, and MANs. Due to this reason it is possible to see a large number of ***heterogeneous components*** in a WAN. WANs use different communication media for interconnection and the nodes can be located as close together as a few blocks or as far away as another country.

Computers connected to a WAN are often connected to a ***Public Network***. They can also be connected through leased lines or satellite links. WAN is mostly used by the government or large organizations because of the huge investments made to implement them.

Categories of Wide Area Networks: - There are three categories of wide area networks, namely: -

Enterprise Network: An interconnected version of all the local area networks of a single organization.

Global Network: A network formed by combining the networks of several organizations over a wide area.

Internet: When two or more independent networks of LAN, BN, MAN or WAN are joined by connecting devices and switching stations they become an Internetwork or Internet. A single authority does not control the internet; the local, national, regional and international authorities control every segment of the internet. Today, most end users who want Internet connection use the services of Internet Service Providers (ISPs). There are International ISPs, National ISPs, Regional ISPs, and Local ISPs.

CARRIERS AND SERVICE PROVIDERS

In a large computer network, interconnected systems are too separated physically and they usually cover a large geographical area spanning cities and countries such that they must use the communication circuits of a registered common carrier (Telephone Company).

In this network, access holders and telephone owners make use of service providers and carriers because these organizations are paid for the services rendered. There are however different types of service providers and carriers and they also offer different services.

Common Carrier is a government-regulated private company that sells or leases communications services and facilities to the public. Common carriers are profit-oriented businesses and their primary purpose is to provide communication circuits and related services for voice, data, and image transmissions. This term is applied most often to United States and Canadian organizations, but sometimes it is used to refer to telecommunication companies (such as government-operated suppliers of telecommunication services e.g. NITEL). Examples of common carriers are Bell Operating Companies, AT&T, MCI, or US Sprint.

Internet Service Provider (ISP) is a company that sells access to the a wide area network (Internet) allowing computer users to send electronic mail (e-mail) and browse the World Wide Web (WWW), among other tasks. In some cases, ISPs sell dial-up access, providing a phone number that lets users dial into the Internet via a computer modem. In other cases, they sell broadband access, a much faster connection to the Internet. Some broadband services, known as digital subscriber lines (DSL) services, run over standard telephone lines, while others use cable television lines or wireless connections to orbiting satellites (e.g. VSAT).

Typically, the ISP must visit the user's home and install the broadband service on the user's computer, but some companies offer toolkits that let users install the service on their own. ISPs typically charge a monthly fee for Internet access, and they often charge an additional fee for broadband installation.

NETWORK TOPOLOGY

Networks topology is the way the devices (nodes) of the network are connected together with links or the geometric arrangement of network computers, which in turn determines the speed of the network and communication efficiency. Its selection depends on the geographic environment, the kind of applications running on the network and the implementation costs. Network topology can be classified in two ways based on: (1). Type of connection (2). Physical topology

TYPE OF CONNECTION

A network is two or more devices connected through links. A link is a communication pathway that transfers data from one device to another. For communication to occur, two devices must be connected in some way to the same link at the same time. There are two possible types of connections.

(1) Point-to-Point connection

(2) Multipoint connection

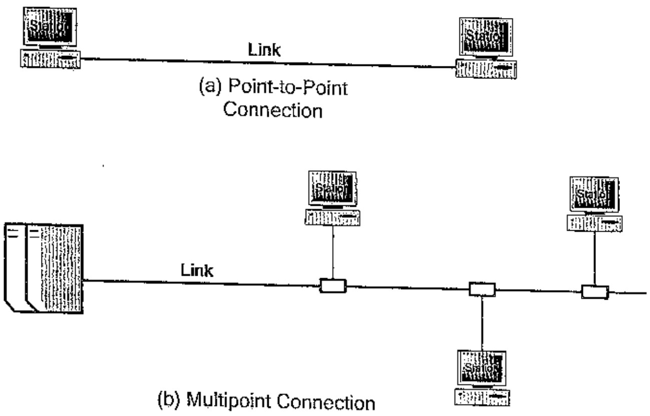


Fig.14: Types of Connection

Point-to-Point Connection provides a dedicated link between two devices. The entire capacity (bandwidth) of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible.

Multipoint (also called **Multidrop**) Connection is one in which more than two devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is spatially shared connection. If users must take turns, it is time-shared connection. Multipoint connections are designed to load the channel more efficiently and thus save money.

PHYSICAL TOPOLOGY

Physical topology refers to the way in which network is arranged physically. Two or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices to one another. There are four basic topologies possible:

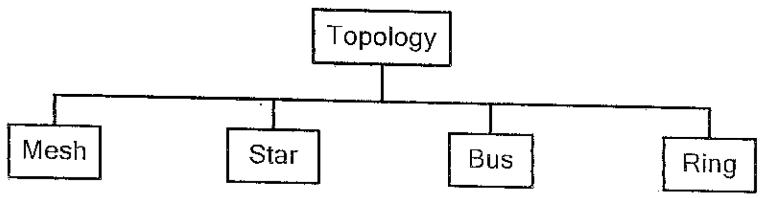
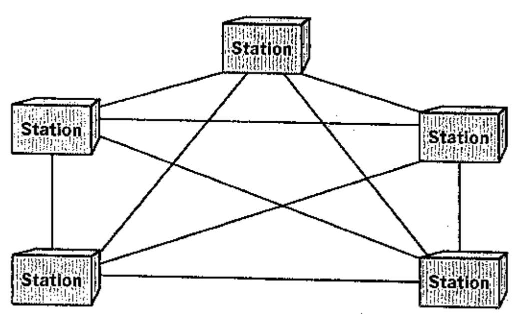


Fig.21: Categories of topology

MESH TOPOLOGY: In this topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. In a mesh network topology, we need full-duplex mode links. One practical example of mesh topology is telephone network.



In a mesh network topology, we need, $\frac{n(n-1)}{2}$ full-duplex mode links.

To accommodate the links, every device on the network must have (n – 1) input/output ports to be connected to the other (n – 1) stations.

Advantages

(1) Use of dedicated links guarantees connection eliminating traffic problems that can occur when links must be shared by multiple devices.

(2) It is robust that is if one link becomes unusable, it does not affect the entire system.

(3) There is the advantage of privacy or security. Because every message travels along a dedicated line, only the intended recipient sees it.

(4) Point-to-point links make fault identification and fault isolation easy. Traffic can be routed to avoid links with suspected problems.

Disadvantages

(1) The amount of cabling and input/output ports required is many because every device must be connected to every other device.

(2) Installation and reconnection is difficult

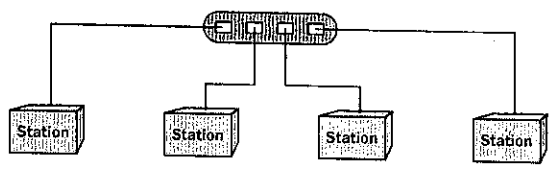
(3) Requires too much space.

(4) Expensive to implement due to the amount of hardware required to connect each link (I/O ports and cable).

STAR TOPOLOGY: Here, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The hub acts as a switch: if one device wants to send data to another, it sends the data to the hub, which then relays the data to the other connected device. The devices are not directly linked to one another. Unlike in mesh topology, a star topology does not allow direct traffic between devices.

The hub usually has multiple ports for multiple links to plug into. All messages must pass through the hub that controls the flow of data. This arrangement makes it easy for the network administration to re-configure the network.

Typical applications of star topology are 10Base-T and 10Base-F Ethernet LAN. The 10BaseT uses UTP wires with RJ45 connectors on both sides of the cable. This cable is used to link a network node to the hub, subject to a maximum distance of 100 metres. 10BaseF uses fibre cable to connect devices to the hub. It can cover greater distances (2000 metres).



Advantages

(1) Less expensive than mesh topology. Each device needs only one link and one I/O port to connect it to any number of others.

(2) Installation and reconnection is easy.

(3) Requires far less cabling

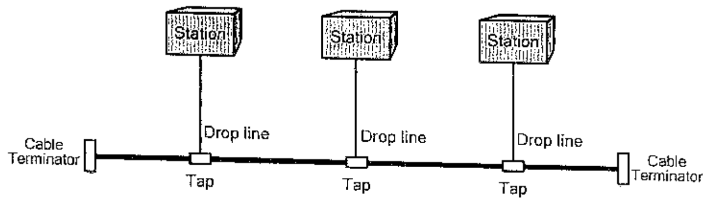
(4) It is robust. If one link fails, only that link is affected. All other links remain active.

(5) Fault identification and isolation is easy.

Disadvantages

(1) The main disadvantage is dependency of the whole topology on one single point the hub. If the hub goes down, the whole system is dead.

BUS TOPOLOGY: This topology uses multipoint connection. One long cable (bus) acts as a backbone to link all the devices in a network. This allows only one device to communicate at a time. Nodes are connected to the bus cable by drop lines and taps. A tap is a T-connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.



Data sent by any node contain source and destination addresses, where each station monitors the bus and copies the data addressed to it. Only one node can transmit at a time. A medium access control protocol (MAC) known as Carrier Sense Multiple Access with Collision Detection (CSMA/CD) determines which station is to transmit.

As a signal travels along the backbone or bus, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between the taps. At the end points of the bus, the terminators absorb the signals to clear up the link.

The major applications of the bus topology are the Thick Ethernet (10Base-5) and Thin Ethernet (10Base-2). The thick Ethernet uses additional device called the transceiver. It can cover a distance of 500 metres per segment and a total of 2500 meters for a maximum of 5 segments with the use of repeaters between segments. Thin Ethernet can cover a distance of 185 metres.

Advantages

(1) Easy to install. Backbone cable can be laid along the most efficient path, and then connected to the nodes by drop lines of various lengths.

(2) Requires far less cabling than mesh and bus topology.

(3) Allows relatively high data rates up to 100 Mbps compared to star and ring topologies.

(4) If a node goes down it does not affect the rest of the network.

Disadvantages

(1) A fault or break in the bus cable stops all transmission, even between devices on the same side of

the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

(2) Difficult reconnection and fault isolation. Once a bus is designed and configured during installation, the configuration is fixed; it can therefore be difficult to add devices.

(3) Signal reflection at the taps can cause degradation in quality. This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable.

(4) Requires a network protocol to detect when two nodes are transmitting at the same time.

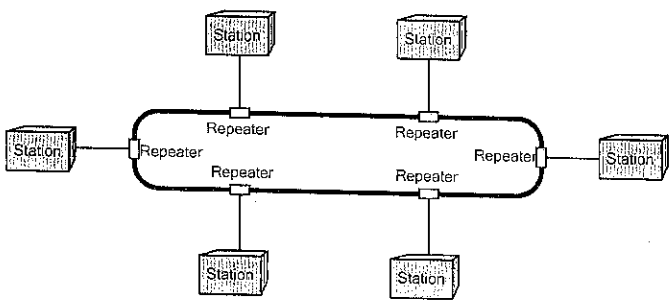
(5) Does not cope well with heavy traffic rates.

RING TOPOLOGY: Here, each computer has a dedicated point-to-point connection with only the two computers on either side of it. Each node is connected to the network via a repeater (active hub) and the repeaters are connected by point-to-point links to form a ring.

A signal (data packet) is passed along the ring in one direction, from node to node, until it reaches its destination. Each node in the ring incorporates a repeater. When a device receives a signal intended for another node, its repeater regenerates the bits and passes them along. If the message reaches the destination node, the repeater at that node accepts the message. The repeater has the ability to recognize its own address in the data packet in order to accept messages.

To allow orderly access to the ring, a single electronic token passes from one computer to the next around the ring. A computer can only transmit data when it captures the token.

A typical application of ring network is IBM Token Ring which uses the Token Ring protocol.



Advantages

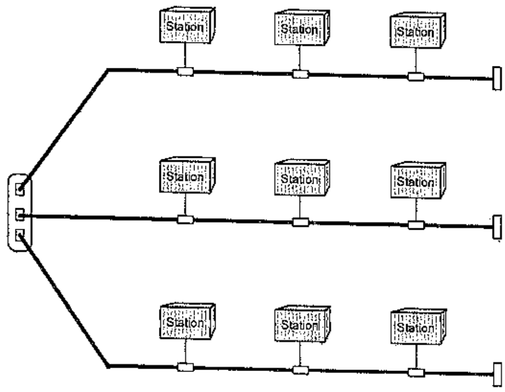
(1) Easy to install and reconfigure.

(2) Fault identification and isolation is easy. Generally in a ring a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

Disadvantages

(1) Unidirectional traffic. In a simple ring a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

HYBRID TOPOLOGY: This is a combination of any two of or more type of topologies i.e. Bus, Star, or Ring topology. Hybrid networks may use a mixture of different types of transmission media (satellite, microwave e.t.c).



EXERCISE

(1). Seventeen workstations are to be arranged in a mesh topology. How many cables are needed? How many ports are needed for each device?

(2). For each of the following four networks, state the consequences if a connection fails.

(a) Five devices arranged in a mesh topology

(b) Five devices arranged in a star topology

(c) Five devices arranged in bus topology

(d) Five devices arranged in ring topology

(3). Sketch a hybrid topology with a ring backbone and 3 bus networks having three stations each.

MODULE 5A
EMERGING TECHNOLOGIES

6.1 INTRODUCTION

Several applications (uses) and solutions to human problems are continuously being developed since the appearance of computer networks. Emerging technologies are the applications that make use of computer networks to facilitate the transfer of information from one place to another in various forms for different purposes. They include the following: -

(1). Teleconferencing
(2). e-banking
(3). e-commerce
(4). Medical Telediagnosis
(5). Electronic Data Interchange (EDI)
(6). Internet Telephony
(7). e-learning
(8). Webcasting

<u>**PERSONAL RESEARCH**</u>

Question: Using online resources (INTERNET) conduct a practical research on all the emerging technologies mentioned above. Your research should be guided by the following sub-topics -

- 1.0 Introduction
- 1.1 Definition, description and diagram
- 1.2 Types the technology that exist
- 1.3 Components or parts of the technology
- 1.4 Method of operation
- 2.0 History of the technology
- 2.1 Development of the technology
- 3.0 Impact on the Public
- 3.1 Benefits (Advantages)
- 3.2 Problems (Disadvantages)
- 4.0 References

<u>**Online Resources:**</u>

(1). www.wikepedia.org
(2). Search engine e.g. www.google.com
(3). www.isoc.org
(4). www.yahoo.com

MODULE 6

NETWORK IMPLEMENTATION AND SECURITY

NETWORKING DEVICES

Networking devices are the hardware components that are used to connect all the computers, workstations, servers and transmission media together to form a functional network. They perform a variety of functions ranging from signal regeneration and media connectivity to advanced functions such as address identification and forwarding and protocol translation. Typical networking devices are: -

(1) Network Interface Card
(2) Passive Hub
(3) Repeater
(4) Active Hub
(5) Bridge
(6) Router
(7) Gateway

These devices are designed to operate at a given layer (or layers) of the OSI Reference Model. Therefore, they can be organized into five different categories based on the layer in which they operate in a network, as shown in fig. 21.

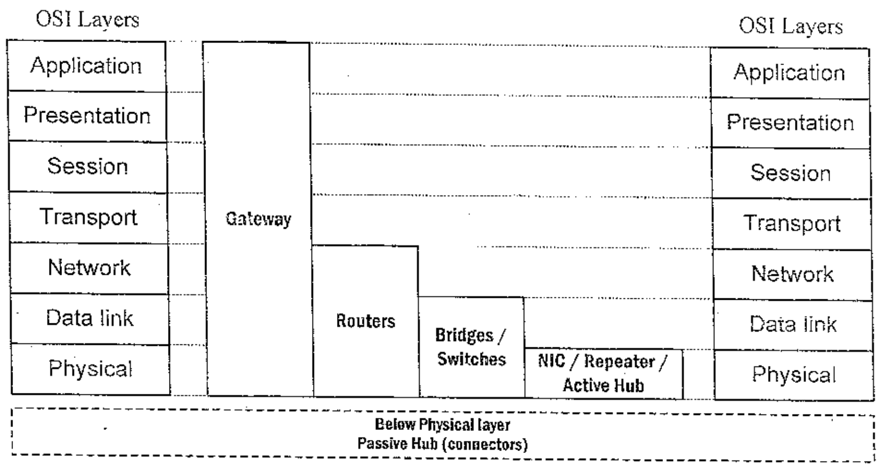


Fig. 21: Categories of networking devices and their layers of operation

The categories are:

(1) Those that operate below the physical layer = passive hub (i.e. connectors)
(2) Those that operate at the physical layer = NIC, Repeater, Active hubs
(3) Those that operate at the physical and data link layer = Bridge, Switches
(4) Those which operate at the physical, data link and network layers = Routers
(5) Those which operate in all seven layers = Gateway

NETWORK INTERFACE CARD (NIC): - This operates at the physical layer. It is sometimes called network adapter or network card or simply NIC. It is the physical connection between a computer/workstation/server and the local area network cable. The physical or MAC address of a computer is located on the NIC.

The NIC card translates data from the computer into electronic/optic signals for transmission in the network medium. It is also responsible for packaging data for transmission, and for controlling access to the network cable. When the data is packaged properly, and the timing is right, the NIC will push the data stream onto the cable.

PASSIVE HUB: - This is a central multi-port connector for connecting cables from several devices in a star configuration. It does not provide an processing or regeneration of signals

This is just a connector. It connects the wires coming from different branches. In order to form a star topology, a hub is used to connect the wires to a common point in the network. It allocates a port to each computer and the computers are interconnected through the hub. It operates below the physical layer.

REPEATERS: - This device operates only at the physical layer of the OSI. It is used to extend the maximum limit on distance over which data can be transmitted on wire or fibre optic cables.

Signals that carry information within a network loses it's strength after travelling some fixed distance on all kinds of media including copper, fibre optic and wireless. This phenomenon known as Signal Attenuation or Signal loss has to be minimized because it corrupts the data that it carries.

A repeater is a device that receives the data signal before it becomes too weak or corrupted and regenerates the original bit pattern and re-transmits it to the next segment. This increases the distance over which a signal can be transmitted.

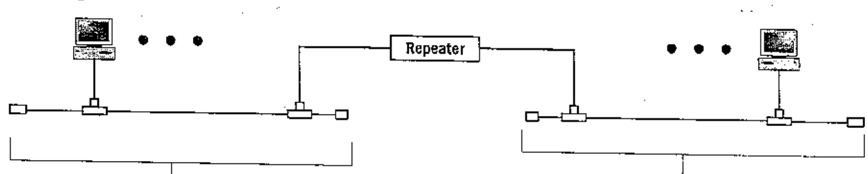
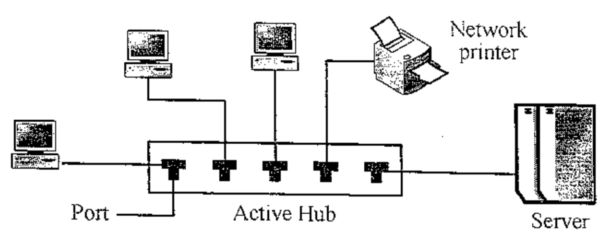


Fig. 22: A repeater connecting two segments of a LAN

ACTIVE HUB: - This is OSI layer-1 hardware to which devices such as servers, workstations, printers and other computers can be connected to form a local area network. Sometimes an active hub is called a multi-port repeater.

Active hubs do not interpret any of the data that flows through them like bridges or switches. They are unaware of the source and destination of packets flowing through them and all the packets flowing through the hub are broadcast to all other devices connected to the hub. Hubs generally have numerous ports so that many network devices can be interconnected.



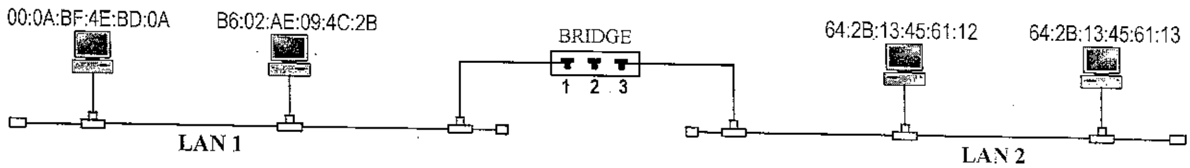
BRIDGE: - A bridge is a LAN device that operates on both physical and data link layer. It connects two or more LAN segments while simultaneously filtering network data transmission between those segments. As a physical layer device it regenerates the signal it receives. As a data link device, the bridge can check the physical (MAC) addresses source and destination contained in a frame.

The MAC address identifies the physical address of the NIC and is different from the network address. A MAC address takes the form of 6-byte address written as 12 hexadecimal digits, such as: 71:2B:13:45:61:41.

The MAC address is set up in the NIC when it is manufactured and cannot be changed. It gives no information on the physical location of the NIC or computer. In LANs, a data frame is transmitted on the network with the destination MAC address.

FILTERING – Filtering is the ability of a bridge to check the destination address of a frame and decide if the frame should be forwarded or dropped; If the frame is to be forwarded the bridge will specify the port. A bridge has a table that links MAC addresses to ports.

Example: In the figure below, two LANs are connected by a bridge. If a frame destined for computer with MAC address = B6:02:AE:09:4c:2B arrives at Port 1, the bridge consults it table to find the departing Port. According to its table a data frame with destination MAC address B6:02:AE:09:4c:2B leaves through Port 1; therefore no need for forwarding and the frame is dropped. On the other hand, if a frame with destination MAC address 64:2B:13:45:61:13 arrives at Port 1, the departing port is Port 3 and the frame is forwarded.



PORT - 1		
MAC address	**Decision**	**LAN segment**
00:0A:BF:4E:BD:0A	Local	Port 1
B6:02:AE:09:4c:2B	Local	Port 1
64:2B:13:45:61:12	Bridge	Port 3
64:2B:13:45:61:13	Bridge	Port 3

In the first case, LAN 2 remains free of traffic; in the second case both LANs have traffic. This example shows only a 3 port bridge in reality a bridge usually has more ports. Note also that a bridge does not change the MAC addresses contained in the frame.

SWITCHES: - Switches are OSI layer-2 devices that evolved from bridge technology. Switches can read frames from multiple ports and create simultaneous forwarding paths.

Switches and bridges have a lot in common. Both can be multiport devices; both have learning algorithm that allow these devices to learn the MAC addresses of all attached network devices; both have filtering ability that allow data transmissions to be forwarded if the data transmission is not intended for a computer on the local segment where the data originated. But there are differences between bridges and switches.

(1) In older networks, bridges were mainly connected to hubs or other bridges. In modern networks a switch port can be directly connected to individual computers or to hubs, bridges, other switches and routers.

(2) A switch is designed to provide a faster performance and increases bandwidth than a bridge. It uses processors that can read multiple ports simultaneously and establish multiple and **simultaneous forwarding paths.**

ROUTING

Routing is the process of determining the route or path through the network that a message will travel from the sending computer to the receiving computer. It refers to sending data from source node to the destination node through many intermediate nodes. As the process requires source and destination address to be attached with the data packet, each packet is transmitted with source and destination address fields. Routing occurs in the Network Layer (Layer 3).

ROUTERS: - A router is an OSI layer-3 device that connects two or more networks and allows data packets to flow between them based on IP addresses and the best possible paths (route). A router normally connects LANs and WANs in the Internet.

To allow data to flow between these networks, each network is assigned its own unique network address that distinguishes it from the other networks and each network is connected to at least one router.

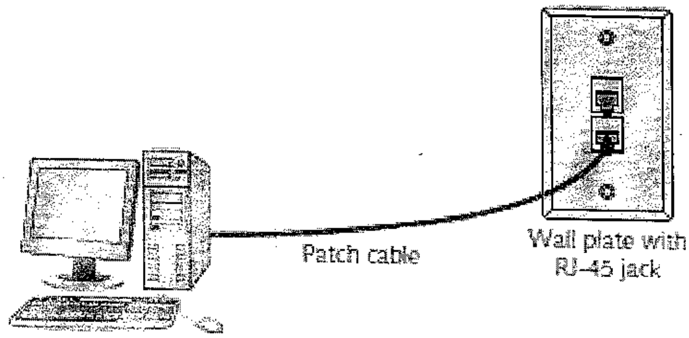
A router has a **routing table** that stores the IP (logical) addresses of other networks so that data from one network can be routed to the appropriate destination network over the best available paths. A router can also function in the physical, data link and network layers. When a router operates at the data link layer it is known as a **Brouter**.

GATEWAYS: - A gateway is hardware or software or combination of both that provides protocol translation or connectivity between dissimilar systems. Gateway can function in all the layers of the OSI model. A gateway can handle data transmission, frame size conversion from one network to the other, it can perform protocol conversions, and it can make data format translations such as from ASCII to EBCDIC. Gateways can also provide security.

STRUCTURED CABLING

This refers to a pre-planned cabling system that is designed with growth and reconfiguration in mind. It specifies how cabling should be organized, regardless of the media type or network architecture. It usually consists of sub-systems. There are six (6) sub-systems that make up a structural cabling system and these are: -

(1) **Work Area:** The work area is where workstations and other user devices are located. It is where people work. Faceplates and wall jacks are installed in the work area, and patch cables connect computers and printers to wall jacks, which are connected to a nearby telecommunications closet.



(2) **Horizontal Wiring:** Horizontal wiring runs from the work area's wall jack to the telecommunications closet and is usually terminated at a patch panel. Acceptable horizontal wiring types include four-pair CAT 5e or CAT 6 or fibre-optic cables. The total maximum distance for horizontal wiring is up to 100 meters, which include the cable running from the wall jack to the patch panel plus all patch cables.

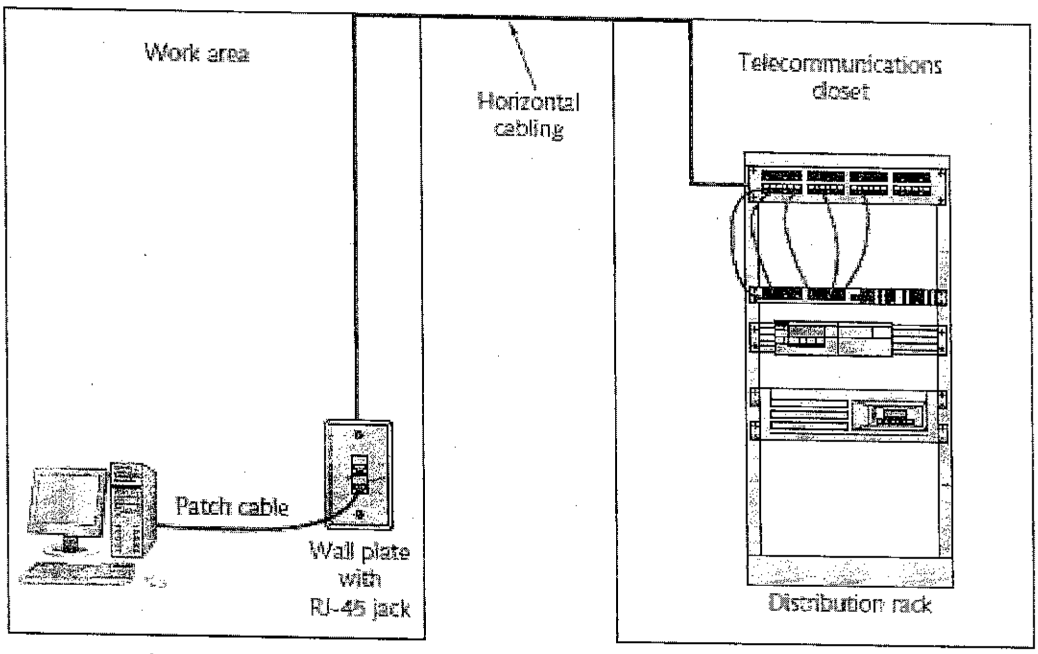


Fig.3 *Work area, horizontal wiring and telecommunications closet*

(3) **Telecommunication Rooms:** The telecommunications room is where the telecom and data communication equipments are installed that connect the backbone horizontal cabling sub-

Same as "Wiring Closet"

systems. It provides connectivity to computer equipment in the nearby work area.

(4) **Equipment Rooms:** The equipment room houses servers, routers, switches and other major network equipment and serves as a connection point for backbone cabling running between telecommunication rooms.

(5) **Backbone Cabling:** Backbone cabling (or vertical cabling) interconnects the telecommunications rooms and equipment rooms. This cabling runs between floors or wings of buildings and between buildings to carry network traffic destined for devices outside the work area. It is often fibre optic cable but can also be UTP if the distance between rooms is less than 90 meters.

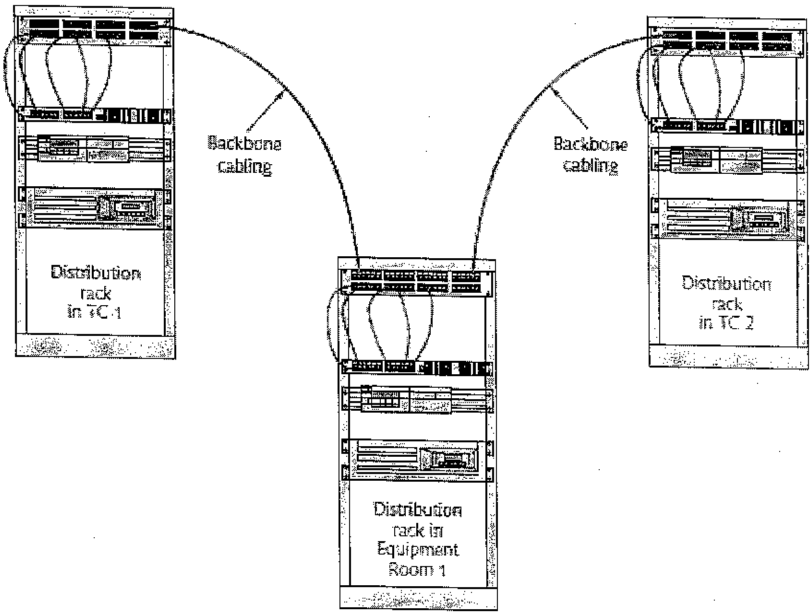


Fig. 5 *Backbone cabling connects Telecommunications Rooms and Equipment Rooms*

(6) **Entrance Facilities:** An entrance facility is the location of the cabling and equipment that connects a corporate network to a third-party telecommunications provider. It can also serve as an equipment room and the main cross-connect for all backbone cabling. This is where the network interfaces the outside world.

NETWORK ADMINISTRATION AND SECURITY

Network Administration is the process of operating, monitoring, and controlling the computer network to ensure it works as intended and provides value to its users. The **network manager** or administrator is responsible for managing all the day-to-day activities of a network to ensure that the network functions properly. The network manager's duties include tasks such as setting up user and groups accounts; making sure that the network is secure; backup and file management; and auditing.

User Accounts: A user is a person who makes use of a computer or network resources. In order for a user to utilize network resources, a user account has to be created for that person. **User Account** is an identity created for a person in a computer system or network which allows potential access to the resources in that system or network. The person is identified to the system by a username or login name. To log into a user account, a person is required to confirm or authenticate oneself personally with a password or other information for the purposes of security, accounting, and resource management.

User Identification involves a range of information used to identify a specific user of a system for security and control purposes.

Authentication is a process whereby a system user is required to provide and verify his or her user identification to gain access to network systems. Normally, after user identification comes authentication, which would require the user to provide additional information unique to that particular user such as password or fingerprint etc. Specific information that can be used to authenticate users are: -

- User name
- Finger prints
- Palm prints
- Passwords
- Retina scans
- Voice prints

Group Account: This account is used to organize users so that allocation of resource permissions and rights can be managed more easily than working with a large number of individual accounts.

NETWORK SECURITY

Computer networks generally share resources used by diverse users and applications representing different interests. The Internet is particularly widely shared, being used by competing businesses, mutually hostile organizations, governments, and opportunistic criminals. Unless security measures are taken, a network conversation or a distributed application may be compromised by an adversary.

NETWORK SECURITY consists of all the provisions and strategies adopted by a network administrator to prevent and monitor unauthorized access, misuse, modification, or denial of a computer network and network-accessible resources. It also deals with the effectiveness of the measures. Security involves both physical and information (data) security.

NEED FOR NETWORK SECURITY

In a networked environment, organizations have become heavily dependent on data communications networks for daily business transactions, database information retrieval, distributed data processing, and internetworking. The advent of the Internet with the possibility of connecting computers anywhere around the world has magnified the need for network security. The goals of network security are to:—

1. Protect network assets namely; hardware, software, information, and services from all types of threats.
2. Minimize the effects of security breaches once they have occurred.

3. Prevent malicious damage to network hardware or files; prevent malicious misuse of hardware and software.
4. Prevent theft of network components or information.
5. Limit accidental damage or destruction of hardware or software, either through user carelessness or environmental events.
6. Protect data confidentiality and integrity.
7. Prevent unauthorized access to a network and unauthorized use of its resources. This goal includes the more specific one of preventing interception or theft of network files or transmissions.
8. Provide for recovery from disasters (fire, flood, theft, and so on). There must be provisions for restoring the network data and getting the network back into service.

There are three basic requirements that must be met for a network security plan to be adequate—

(1) Confidentiality
(2) Integrity
(3) Availability

Availability: This means that the network components, information, and services are available whenever needed.

Confidentiality: Services and information are available only to those authorized to use them. The contents of a message when transmitted across a network must remain confidential, i.e. only the intended receiver and no one else should be able to read the message. The users, therefore, want to encrypt the messages they send so that an eavesdropper on the network will not be able to read the contents of the message.

Integrity: Components and information are not destroyed, corrupted, or stolen, either through outside intervention or through in-house incompetence. The contents of a message when transmitted across a network must reach the destination without alteration. If any alteration, malicious or accidental incident occurs during transmission, the receiver should be able to conclude that an alteration has happened.

SECURITY THREATS

A **security threat** to the data communication network may be defined as any potential adverse event that breaks one or more security goals. For example, losses of hardware or data are threats to a network's security, as are thefts of passwords or user identification (ID). The security of a network can be threatened or breached with respect to four aspects—

1. Hardware,
2. Software,
3. Information,
4. Network operation.

The following are some of the threats that can breach a network's security—

Interruption	Theft	Corruption	Dishonesty
Destruction	Deletion	Bugs	Incompetence
Damage	Interference	Loss	Denial of service
Unauthorized access/use	Disaster	Overload	Power anomaly

SECURITY MEASURES

The following steps or actions should be taken in the planning and implementation of a security plan for a network.

Physical Security: - Involves all the measures taken to deny physical access to areas containing sensitive information. If access to is prevented to physical components of the network such as terminals, CPU, hard disks, modems and communication links, the likelihood of unauthorized access is significantly decreased. Measures taken to implement physical security are,

- ⇨ Door locks
- ⇨ Safes
- ⇨ Security guards

Data Security: - Involves all the measures taken to protect organizational information from security threats due to disruptions, destruction, disaster, and unauthorized access. These measures involve –

- ⇨ Backups
- ⇨ Redundant systems
- ⇨ Firewalls
- ⇨ Cryptography
- ⇨ Uninterruptible Power Supplies

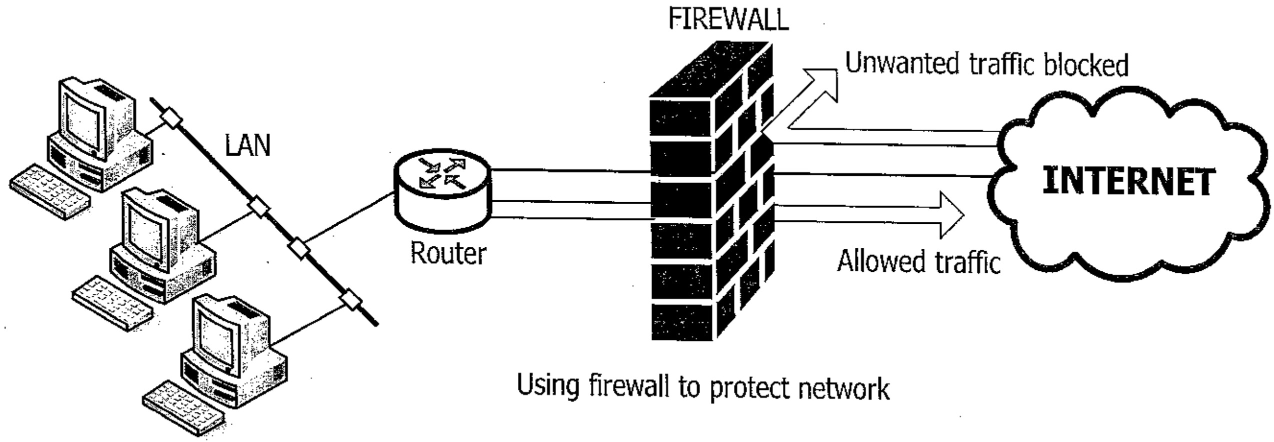
Backups: - A backup is an archival copy of computer information that is stored on an external medium. For example, a backup might contain the contents of a hard disk or a directory. The creation of regular backups is essential in a networking environment. An effective backup system ensures that data stored on the network can be recreated in the event of a crash or another system failure. Backups are generally made to tapes, optical disks (CD, DVD), or external hard disks.

Redundant Systems: - This is a network configuration with extra systems or components built into the network as a means to prevent network total breakdown. The redundant components are included to make it possible to compensate for malfunctions or errors. Redundancy may be applied to hardware, software, or information. Duplicate hard disks, servers, or cables are examples of hardware redundancy that provide data security.

Network Security can be implemented at multiple levels within a network. There are four measures that can be implemented at various levels in the network: -

(1) Identification and Authentication
(2) Firewalling
(3) Intrusion Prevention System (IPS)
(4) Cryptography

Firewall: A firewall is a hardware device or software program installed between a network and the Internet that checks data traffic going into or out of the network or computer, in order to block or allow these data packets based on a set of rules. This provides a security barrier to the internal network from external network or Internet. It protects the internal network from attack, or unauthorized penetration, by outside invaders. A firewall may be a specially programmed router or software installed in the operating system of a computer that checks data traffic flowing into and out of the network or computer. This is illustrated below.



Using firewall to protect network

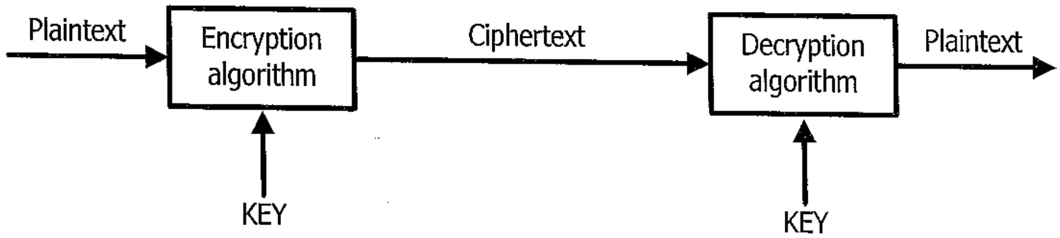
Intrusion Prevention System (IPS): These are systems designed to monitor all activities in a computer network for signs of unauthorized access and to take appropriate actions to stop it. The actions include reporting, blocking, or dropping it when it occurs.
Intruders can be organization employees or external hackers who steal data (e.g., customer credit card numbers) or destroy important records. A security policy defines the key stakeholders and their roles, including what users can and cannot do.

Cryptography: This is a process whereby transmitted message is scrambled (disguised) by the use of mathematical rules known as algorithms at the sending location and reconstructed into a readable message at the receiving end. This makes them secure and protected from any network threat.

Encryption is the process of disguising the message to be transmitted, whereas **decryption** is the process of restoring it to readable form. When the message is in readable form, it is called **plaintext**; when in encrypted form, it is called **ciphertext**. Encryption can be used to encrypt files stored on a computer or to encrypt data in transit between computers. Message encryption and decryption involves use of its two components:—

(1) Algorithm (mathematical rule) for both encryption and decryption.
(2) Unique key that the algorithm uses with the plaintext for encryption or decryption.

Encryption algorithm is the computer program that converts plaintext into ciphertext. The ciphertext is the message after the encryption algorithm has been applied. A **KEY** is the unique piece of information that is used to create ciphertext and then to decrypt the ciphertext back into plaintext. After the ciphertext is created, it is transmitted to the receiver, where the ciphertext data is decrypted by the decryption algorithm which is the same program as the encryption algorithm but this time it converts the ciphertext back into plaintext.



Encryption and decryption using keys

The encryption or decryption algorithm may use the same key or different keys. The algorithm is generally well known but the key is always kept secret. There are of two types of algorithms that encrypt a message:—

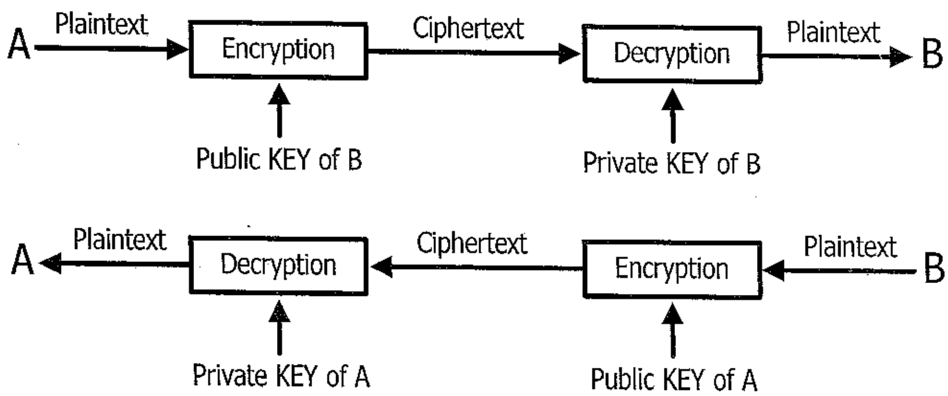
(1) Secret-key encryption (2) Public-key encryption

Secret-key encryption algorithm is also known as **symmetric-key** encryption. It allows the sender and receiver to share the same key for encryption and decryption. Examples are:—

(1) Data Encryption Standard (DES)
(2) Advanced Encryption Standard (AES)
(3) Triple DES
(4) International Data Encryption Algorithm (IDEA)

Public-key encryption algorithm is also known as **asymmetric-key** encryption. In this case, the encryption and decryption keys are different. Each user has two keys— public key and private key. The public key is used for encryption and the private key is used for decryption. Decryption cannot be done using the public key. The two keys are linked but the private key cannot be derived from the public key.

The public key is well known but the private key is secret and known only to the user who owns the key. In other words, everybody can send a message to the user using his (user's) public key. But the user only can decipher the message using his private key. This is shown below.



Encryption and decryption using public and private keys

Examples of public key encryption algorithms are:

(1) RSA (2) El Gamal (3) Digital Signature Standard (DSS).